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South Pacific albacore: manuscript workflow example

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1 Executive summary

This is an editable manuscript workflow example, not a stock assessment. The year 2027 demonstrates a future writing project. Diagnostic and stepwise outputs use archived ALB fits; other studies include illustrative display values. No new assessment or management conclusion is supplied. Replace the example results and bracketed writing prompts before using this document as an assessment.

This paper describes the 2027 stock assessment of South Pacific albacore tuna (*Thunnus alalunga*) in the South Pacific Ocean.

[Summarise the data period, changes since the previous assessment, principal sources of uncertainty and evidence supporting the stock-status conclusions.]

[State the assessment area and the periods and denominators used for recent spawning potential, depletion and fishing mortality. Report the retained model ensemble and relevant uncertainty intervals.]

[Summarise management implications, limitations and priorities for research.]

2 Introduction

South Pacific albacore supports longline fisheries throughout the South Pacific and surface fisheries in southern waters. Stock assessments integrate information on fishery removals, relative abundance, size composition and biology to estimate population dynamics and evaluate stock status.

The spatial scope and structure of previous assessments have evolved with available data and assessment needs. The 2012 assessment considered the South Pacific using an areas-as-fleets approach (Hoyle et al., 2012). Subsequent assessments introduced explicit spatial structure within the Western and Central Pacific Fisheries Commission Convention Area (WCPFC-CA) (Harley et al., 2015; Tremblay-Boyer et al., 2018). Comparisons between assessments therefore require consistent geographical coverage, fishery definitions and reference-point calculations.

[Describe the most recent assessment, subsequent scientific review and the objectives of this update. Add the current assessment and supporting-paper citations; the historical references above are not a complete assessment history.]

Integrated assessment models combine population and fishery dynamics with observation models for the available data. MULTIFAN-CL provides one established implementation of this approach (Fournier et al., 1998; Hampton and Fournier, 2001; Kleiber et al., 2019). The framework, version and model components used in this assessment are described in Section 5. Each assessment re-estimates the population history using the available data and assumptions. Changes in historical estimates can therefore reflect both new observations and changes in model structure.

Uncertainty within a fitted model and uncertainty about model structure are distinct. Sensitivity analyses examine the effects of individual assumptions; an ensemble can represent a specified set of plausible alternatives. The construction, weighting and numerical acceptance of that ensemble must be considered when interpreting its distribution of stock-status estimates.

3 Background

3.1 Stock structure and movement

Albacore occur throughout tropical and temperate oceans. North and South Pacific albacore are generally treated as separate stocks, although population structure and connectivity within the South Pacific remain subjects of investigation (Anderson et al., 2019; Nikolic et al., 2017). Evidence relevant to stock structure includes tagging, spatial variation in length composition, growth, genetics and the seasonal distribution of fisheries.

Tagging studies and size-composition patterns indicate movement between southern feeding areas and warmer waters as albacore grow. Juveniles are prevalent in cooler waters, whereas mature fish occur farther north during the spawning season (Farley et al., 2014; Nikolic et al., 2017). Historical tag releases and recoveries have uneven geographical and temporal coverage. Their interpretation must account for the distribution of release activity, fishing effort and reporting.

Biological evidence from tagging does not imply that tag-recapture observations are included in the assessment likelihood.

Longitudinal variation in growth and biological characteristics provides further information on population structure (Anderson et al., 2019; Macdonald et al., 2013; Williams et al., 2012). Smaller fish in parts of the eastern Pacific and spatial patterns in length composition also inform fishery stratification (Vidal et al., 2021). These observations need not imply closed populations at assessment-region boundaries.

Seasonal catch patterns have been used to investigate the distribution and movement of albacore (Langley, 2004). Habitat-based ecosystem models provide another source of hypotheses about recruitment distribution and movement (Lehodey et al., 2015; Senina et al., 2020). Such hypotheses should be distinguished from movement estimated from the assessment data.

3.2 Vertical behaviour

Albacore occupy a broad depth range, and their availability to fishing gears depends on temperature, prey and vertical behaviour. Longline catches in warmer waters commonly occur at greater depths than catches in cooler waters, whereas troll fisheries operate near the surface (Bigelow et al., 2006). Electronic tagging of large albacore around New Caledonia, Tonga and New Zealand documented variation in depth use, including deeper daytime and shallower nighttime habitat use (Williams et al., 2015). These patterns are relevant to interpreting catchability and changes in the composition of longline catches.

3.3 Biology

3.3.1 Reproduction

Spawning occurs in warm waters of the South Pacific during the austral summer, with reproductive studies documenting latitudinal and seasonal variation (Farley et al., 2013b; Farley et al., 2014). Estimates of maturity differ with the biological definition, sampling design and treatment of spatial variation. The maturity ogive and sex-ratio assumptions used in the assessment should therefore be distinguished from the range of estimates reported in the literature.

Reproductive potential describes the relative contribution of fish of different ages or sizes to reproduction. Depending on the model, it may combine maturity, female proportion, body mass and other reproductive information. A spawning potential index based on a scaled reproductive ogive is not necessarily spawning biomass measured in tonnes. The definition and units used here are specified in [Section 5.2.4](#).

3.3.2 Growth

Young albacore move into southern surface fisheries at approximately one year of age. Length-frequency distributions from troll fisheries and ages estimated from otoliths provide complementary information about growth (Farley et al., 2013a; Williams et al., 2012). Growth slows with age, and large albacore can attain lengths exceeding 100 cm. Otolith studies have recorded ages

up to approximately 15 years (Farley et al., 2013a). These observations describe sampled fish and do not prescribe the model’s maximum age or plus group.

Growth differs between sexes and across the South Pacific (Farley et al., 2013a; Williams et al., 2012). Ageing methods, within-year timing and the treatment of the youngest fish can also influence the estimated growth curve. Reanalysis of otolith ages was considered in the assessment work reported by Farley et al. (2021). Growth hypotheses derived from age data and from length-frequency modes can be compared, while recognising their different sampling processes and potential sources of bias.

3.3.3 Natural mortality

Natural mortality is a substantial source of uncertainty for albacore assessments. Previous assessments have used alternative mortality assumptions (Harley et al., 2015; Tremblay-Boyer et al., 2018). Life-history information and empirical relationships between mortality and body size can inform age-dependent hypotheses (Lorenzen, 1996). These relationships are assumptions about mortality, rather than direct observations of mortality in the assessed stock. Growth and mortality hypotheses may be linked and should be varied coherently when they use the same biological inputs.

3.4 Fisheries

South Pacific albacore are caught principally by longline fleets, with surface fisheries targeting smaller fish in southern waters. Longline fleets include distant-water fleets and fleets operating from Pacific Island countries and territories. Troll fisheries operate around New Zealand and in the subtropical convergence zone. Historical driftnet fisheries also contributed to removals and must be represented where they fall within the model period (Nikolic et al., 2017; Vidal et al., 2021).

[Update the description of current fleets, fishing grounds, seasonality and catch trends from the data used in this assessment. Identify the reporting period and the treatment of incomplete recent catch records.]

4 Data compilation

4.1 Spatial and temporal stratification

Spatial stratification should reflect the population processes, fishery operations, data coverage and geographical scope of the assessment. Explicit population regions and areas used to stratify fisheries are different: an areas-as-fleets formulation can retain spatial differences among fisheries without representing separate population states (Waterhouse et al., 2014).

[Specify the assessment boundary, population regions, fishery strata and any aggregation to reporting areas. Describe changes from the previous assessment and add the generated region-map reference.]

[Give the first and last model years, model time step, data extraction date and treatment of incomplete terminal-year records. Explain aggregation used for annual displays when the fitted model uses shorter time steps.]

4.2 Fishery definitions and removals

Fisheries are defined to group observations with reasonably consistent selectivity and catchability. Gear, fishing method, flag, area and changes in operations may inform those definitions. Extraction fisheries account for removals; abundance indices provide information about relative abundance. Keeping these roles explicit avoids attributing removals to an index solely because of its implementation in a particular model engine.

[Describe fleet definitions, catch units, conversion factors, missing-catch treatment and data revisions. Reference the generated fishery-definition table and annual catch figures. Describe observed catch totals without implying that conditioned catches are independent measures of model fit.]

4.3 Abundance indices

Standardised catch per unit effort (CPUE) can provide an index of relative abundance after accounting for changes in fishing operations and sampling. Spatiotemporal approaches can account for spatial dependence and integrate predictions over a specified area (Thorson et al., 2015). Previous albacore analyses combined longline fleets to improve spatial and temporal coverage (Vidal et al., 2021). Covariate effects, changes in fleet coverage and index scaling remain relevant to interpretation.

[Describe the current standardisation, fleets, covariates, prediction domain, index units and uncertainty. State whether observation CVs are fixed, rescaled or estimated; give the rule and its source. Document any regional scaling and the treatment of early observations where fleet coverage differs.]

4.4 Length composition

Length-frequency samples come from observer, port-sampling and other fishery programmes. Sampling coverage can vary among fleets, regions and periods, and fish measured within a sampling event may not be independent. Sampling design and selection of fish for measurement therefore matter for both composition proportions and their likelihood weights.

[Specify fork-length units, bin edges, sampling units, pooling, expansion, minimum sample criteria and handling of missing or zero bins. Distinguish the raw number measured from the effective sample size used in the likelihood.]

4.5 Age and growth observations

Age-length observations can inform growth externally or enter the assessment through an observation model. A forward age-length key describes $P(L | A)$; a conditional age-at-length

model describes $P(A | L)$. The latter depends on the age distribution and sampling process as well as on growth. In general,

$$P(A = a | L = l) = \frac{P(L = l | A = a) \pi_a}{\sum_b P(L = l | A = b) \pi_b},$$

where π_a is the age distribution appropriate to the sampled population, including selectivity when required. Reversing the conditioning by transposing a matrix does not give the same quantity.

[Describe the ageing protocol, age-reading error, sex and spatial coverage, sampling design and how these observations are used. Remove this subsection if no age data inform the assessment.]

5 Model description

5.1 General characteristics

An integrated model links population dynamics, fishing processes and observation models. The following subsections describe the diagnostic model; alternatives are documented in Section 6. Technical descriptions of MULTIFAN-CL provide background for models using that formulation (Hampton and Fournier, 2001; Kleiber et al., 2019).

[State the model framework and version, data and configuration identifiers, spatial structure, time step, age classes and plus-group treatment. Identify estimated parameters, fixed quantities and any random effects.]

5.2 Population dynamics

5.2.1 Recruitment and initial population

Recruitment is the entry of fish into the youngest modelled age class. Its timing, seasonal distribution and allocation among regions must be defined consistently with that age class. Recruitment variation may be estimated around a mean or a stock-recruitment relationship.

For a Beverton-Holt relationship, steepness is recruitment at 20% of unfished equilibrium spawning potential divided by unfished equilibrium recruitment (Francis, 1992). Information about steepness can be limited, so fixed values and alternative hypotheses require an explicit rationale.

[Describe the recruitment parameterisation, deviation variance, correlation, bias correction, terminal constraints and spatial allocation. Give the initial population assumption and any estimated initial age-structure deviations.]

5.2.2 Growth and natural mortality

Growth links the age-structured population to observed length composition and, with a weight-length relationship, to biomass. Its mean and variability must both be specified. Where biological

relationships are fixed externally, their uncertainty may be explored through alternative model inputs.

[Describe the growth function, length-at-age distribution, age-reading error, weight-length conversion and any sex, region or time effects. Give natural mortality in its stated time unit and document conversion to the model time step. Explain any coupling between growth and mortality hypotheses.]

5.2.3 Movement

Movement connects regional population states. Its interpretation depends on the transition model and the ordering of movement, recruitment, fishing and natural mortality within a time step. Transition probabilities are different from instantaneous movement rates.

[For a spatial model, describe allowed connections, seasonal and age effects, estimation or external inputs, and constraints. Remove this subsection for a pooled population.]

5.2.4 Reproductive potential

[Define spawning potential from abundance and the reproductive ogive. State whether the ogive includes body mass, maturity, female proportion or fecundity, and give the resulting units. Describe any conversion from length to age and any normalisation. Do not label a relative spawning-potential index as tonnes.]

5.3 Fishery dynamics

Selectivity describes the relative susceptibility of fish of different ages or lengths to a fishery. Catchability scales an abundance index, or relates effort to fishing mortality when effort is part of the fishing model. Changes in one process can be confounded with changes in another.

[Describe each selectivity form, parameter sharing, time variation and constraints. State how index catchability is estimated and how the model determines fishing mortality. For catch conditioning, describe the catch equation, solver, fishing-mortality bound and achieved catch tolerance.]

5.4 Observation models and weighting

[Identify each likelihood contribution and its assumptions. For lognormal indices, specify the observation-error scale and whether predictions represent an arithmetic mean or a median. Distinguish a fixed catch constraint from a high-precision catch likelihood.]

Length-composition weights represent the information assigned to samples; they need not equal the number of fish measured. The likelihood and its weighting must be documented together. Multinomial, Dirichlet-multinomial and robust normal likelihoods use different variance assumptions.

For the MFCL-compatible robust length-composition formulation, the effective sample size of original record i in fishery f is $n_{\text{eff},i} = \min(n_{\text{raw},i}, 1000)/d_f$, where d_f is the fishery-specific

divisor. For observed proportions o_b , predictions p_b , B bins and effective sample size n , the robust contribution is

$$v_b = \frac{o_b(1 - o_b) + \epsilon/B}{n}, \quad \ell = - \sum_{b=1}^B \log \left[\exp \left\{ - \frac{(o_b - p_b)^2}{2v_b} \right\} + 0.001 \right].$$

[Record the actual divisors, ϵ and the setting controlling normalising constants. If samples are pooled annually, state separately how proportions and effective sample sizes are combined. Raw-count pooled proportions with summed record-level effective sample sizes define an approximation; a single annual robust likelihood is not identical to the original quarterly likelihood.]

5.5 Estimation and model evaluation

Parameters are estimated by minimising the negative log-likelihood plus the specified penalties. Automatic differentiation can provide derivatives for numerical optimisation (Fournier et al., 2012). A small gradient is a numerical criterion and does not by itself establish model adequacy or identifiability.

[Specify fitting phases, initialisation, bounds and convergence criteria. Describe starting-value trials and the use of Hessian diagnostics, parameter correlations, residuals and simulation tests. Report which uncertainty calculations are supported by the fitted model.]

6 Model runs

6.1 Developments since the previous assessment

[Describe changes in data, model engine, spatial structure and assumptions in their actual development order. Identify the starting model and each subsequent step. Discuss changes in both population estimates and fits to individual data sources, using the stepwise figures and interactive comparison.]

6.2 Sensitivity analyses

Albacore assessment sensitivities may include growth, natural mortality, recruitment, movement, index treatment and composition weighting. The choice depends on the current data and model. A single-change sensitivity isolates an assumption relative to the diagnostic model, whereas combinations of changes can reveal interactions.

[Describe the selected alternatives and their scientific rationale. State which parameters are re-estimated. Omit assumptions that are absent from the model.]

6.3 Structural and estimation uncertainty

[Define the ensemble design, model weights, numerical retention criteria and handling of excluded fits. Distinguish variation among central model estimates from estimation uncertainty within models. State whether the reported intervals include both sources and how they are combined.]

7 Model diagnostics

7.1 Starting-value trials (jitter)

Starting-value trials evaluate sensitivity to numerical initialisation by refitting the same data and model from perturbed parameter values. Comparisons of final objectives, gradients and derived quantities help identify alternative solutions and fits that have not reached the same numerical optimum.

[Describe the perturbation scale, parameters perturbed, number of trials and acceptance criteria. Report the retained trials and compare their objectives and population estimates with the diagnostic fit. Link the jitter figures and interactive comparison.]

7.2 Simulation-estimation tests (self-test)

Simulation-estimation tests generate observations under a specified fitted model and refit the model to those observations. Recovery of generating parameters and population quantities assesses the behaviour of the estimation procedure under those assumptions. Bias, dispersion and interval coverage can be examined separately.

[Specify the generating model, simulated data components, treatment of process variation, number of replicates and refitting procedure. Report recovery and convergence, including how failed fits were handled. Distinguish conditional recovery under the generating model from robustness to alternative assumptions.]

7.3 Retrospective analysis

Retrospective analyses successively remove terminal data and refit the model. Estimates from each peel are compared with the full-data fit over their shared periods (Cadigan and Farrell, 2005). Consistent shifts in recent spawning potential, depletion, recruitment or fishing mortality can reveal retrospective patterns.

[State the number and length of peels and which data were truncated. Compare the trajectories and report the definition and period of any summary statistic, such as Mohn's rho. Discuss systematic changes and uncertainty in terminal years.]

7.4 Hindcast evaluation

A hindcast evaluates predictions against observations withheld from fitting. For each training cutoff, the model uses only information available through that cutoff. Predictions for later observations are then compared with the withheld values. This evaluates predictive performance rather than revisions to estimates within the fitted period.

[Specify the cutoffs, forecast horizons, predicted data types, future covariates and baseline forecast. Show predictions with withheld observations by series. Report bias and prediction error on a stated scale, and uncertainty coverage when intervals are available. Any scaling or baseline error used for skill scores must be estimated from the training data.]

7.5 Likelihood profiles and identifiability

Likelihood profiles re-optimize other parameters while a focal parameter or derived quantity is fixed across a range. Total average biomass can serve as a population-scale quantity for MFCL-compatible models (McKechnie et al., 2017; Tremblay-Boyer et al., 2017). Contributions from individual data sources help identify which observations inform population scale and where sources conflict. A penalised-objective profile must be distinguished from a pure likelihood profile when interpreting likelihood-ratio thresholds.

[Define the profiled quantity, averaging period, units, constraint implementation and convergence criteria. Discuss profile shape, parameter bounds and Hessian diagnostics. State which uncertainty calculations are supported by the fit.]

8 Results

8.1 Consequences of model developments

[Summarise the changes in population scale, trends and data fits across the stepwise sequence. Identify consequential interactions rather than repeating every model setting.]

8.2 Model fit

8.2.1 Abundance indices

[Evaluate observed and predicted CPUE by index and region, including early periods with limited fleet coverage. Discuss residual patterns and uncertainty.]

8.2.2 Length and age composition

[Evaluate pooled length-frequency fits and changes through time by fishery. Discuss systematic size-dependent residuals, growth information and differences among fleets. Include age-data fit only where those data are fitted.]

The numerical, simulation and predictive evaluations are described in [Section 7](#).

8.3 Estimated population and fishery processes

[Interpret estimated selectivity and, for spatial models, movement. Describe recruitment, spawning potential, depletion and fishing mortality by region and overall. Explain how uncertainty affects interpretation of recent years.]

8.4 Sensitivity and ensemble results

[Identify the assumptions with the greatest influence on stock status and population scale. Summarise the retained ensemble without treating equally weighted structural alternatives as a posterior sample unless justified.]

9 Stock status and interpretation

9.1 Depletion and fishery impact

Dynamic spawning depletion compares spawning potential under fitted fishing, SB_t , with a corresponding no-fishing counterfactual, $SB_{t,F=0}$:

$$D_t = \frac{SB_t}{SB_{t,F=0}}.$$

The counterfactual retains specified model processes while removing fishing. Its recruitment treatment must be stated, including any stock-recruitment feedback. This comparison separates fishing effects from variation represented in both trajectories; it differs from dividing spawning potential by its initial or equilibrium value.

A management depletion measure may instead divide mean spawning potential over a specified recent period by mean no-fishing spawning potential over a different reference period. That measure is distinct from D_t . Similarly, the mean of quarterly depletion ratios need not equal the ratio of annual mean spawning potentials.

[Define the exact numerator, denominator, spatial domain and years used for each reported quantity. Use these definitions consistently in prose, figures, tables and projections. For fishery impacts, describe the counterfactual removal of individual fishery groups and the attribution method.]

9.2 Yield and reference points

Equilibrium yield analysis evaluates sustainable yield over a range of fishing mortality multipliers under a specified age-specific fishing pattern, stock-recruitment relationship and biological assumptions. The maximum defines MSY ; associated quantities include F_{MSY} and SB_{MSY} (Kleiber et al., 2019). These equilibrium quantities depend on the selected fishing pattern and need not be valid for separate regions of a connected stock.

[State the recent fishing period, recruitment assumptions, regional scope and search procedure. Specify the currently applicable limit and target reference points and cite their adopted definitions. Do not carry forward an old target or averaging period without checking its applicability.]

9.3 Stock status

Kobe plots compare F/F_{MSY} and SB/SB_{MSY} . Majuro plots compare fishing mortality relative to F_{MSY} with a defined depletion measure. Both require explicit temporal and spatial definitions; probabilities derived from an ensemble also depend on model weighting and the representation of uncertainty.

[Present the current stock-status estimates and uncertainty. Interpret the relevant limits and targets only for the geographical area to which they apply.]

10 Projections

Projections extend the population model under specified fishing and recruitment assumptions. Catch, fishing mortality and effort scenarios imply different controls and must be labelled accordingly. Recruitment variation, parameter uncertainty and variation among models contribute different sources of projection uncertainty.

10.1 Scenarios and methods

[Specify the projection start, horizon, controls, recruitment resampling or process model, ensemble weights and uncertainty intervals. Use the same reference-point definitions as the stock-status analysis. Document infeasible catch targets and the treatment of boundary years.]

10.2 Projected outcomes

[Compare projected spawning potential, depletion, catches and relevant reference-point probabilities among scenarios. Describe which sources of uncertainty are included and the assumptions on which the comparison depends.]

11 Discussion

11.1 Assessment interpretation

[Discuss what the assessment establishes about the stock, the evidence driving population scale and trends, and the implications for management. Compare previous assessments using compatible spatial and reference-point definitions.]

11.2 Data and model limitations

[Discuss the main unresolved uncertainties, including index representativeness, composition sampling, growth, mortality or movement where relevant. Distinguish limitations demonstrated by diagnostics from hypotheses requiring further work.]

11.3 Conclusions and research priorities

[State the conclusions supported by the current results. Identify practical priorities for data collection, biological research and model development.]

12 Acknowledgements

[Acknowledge data providers, national fisheries agencies, sampling and ageing programmes, scientific contributors and reviewers involved in this assessment.]

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14 Tables

Table 1: Public report, interactive supplements and published outputs for this workflow demonstration.

Resource	Figures and tables	Interactive output
Workflow demonstration	PDF report	HTML report
Diagnostic fit	HTML supplement	Model viewer
Stepwise development	HTML supplement	Compare development steps
Jitter	HTML supplement	Model viewer
Self-test	HTML supplement	Model viewer
Retrospective analysis	HTML supplement	Model viewer
Sensitivities	HTML supplement	Model viewer
Uncertainty ensemble	HTML supplement	Model viewer
Objective profiles	HTML supplement	Model viewer
Additional output examples	HTML supplement	Model viewer
Hindcasting	HTML supplement	Model viewer
Projection scenarios	HTML supplement	Model viewer
Published outputs	Public repository	Rendered artifacts

Table 2: Fishery definitions and observation units.

Fishery	Region	Role	First year	Last year	Unit
1	1	Catch	1954	2022	fish
2	1	Catch	1954	2022	fish
3	1	Catch	1954	2022	fish
4	1	Catch	1954	2022	fish
5	1	Catch	1992	2022	fish
6	1	Catch	1982	2022	fish
7	1	Catch	1987	2022	fish
8	1	Catch	1954	2022	fish
9	1	Catch	1984	2022	fish
10	1	Catch	1985	2022	fish
11	1	Catch	1982	2022	t

Fishery	Region	Role	First year	Last year	Unit
12	1	Catch	1987	2022	t
13	1	Catch	1983	1990	t
14	2	Catch	1957	2022	fish
15	2	Catch	1961	2021	fish
16	2	Catch	1963	2021	fish
17	2	Catch	1986	2006	t
18	1	Index	1954	2022	relative index
19	1	Index	1992	2022	relative index
20	2	Index	1954	2022	relative index

Table 3: Management quantities and supplied uncertainty summaries.

Quantity	Period	Median	50% interval	80% interval	95% interval
$SB/SB_{F=0}$	2016–2020	0.45	0.41–0.49	0.37–0.53	0.33–0.57
F/F_{MSY}	2016–2020	0.68	0.61–0.75	0.54–0.82	0.46–0.90
SB/SB_{MSY}	2016–2020	1.35	1.22–1.48	1.08–1.62	0.94–1.76

Table 4: Stock-status probabilities for the declared reference points.

Condition	Probability
$SB/SB_{F=0} < 0.20$	0.01
$F/F_{MSY} > 1$	0.05
$SB/SB_{MSY} < 1$	0.04

Table 5: Model reference points and their definitions.

Quantity	Median	Lower	Upper	Unit
MSY	60000	48000	72000	t/year
F_{MSY}	0.3	0.24	0.36	per year
SB_{MSY}	1000000	800000	1200000	t
SB_{MSY}/SB_0	0.28	0.22	0.34	ratio

Table 6: Illustrative projected quantities in the final projection year; medians and central 80% intervals.

Scenario	Quantity	Year	Median	Lower	Upper	Unit
fixed-catch	biomass	2052	0.9499	0.9473	0.952	ratio
	depletion					

Scenario	Quantity	Year	Median	Lower	Upper	Unit
fixed-catch	catch	2052	45150	45150	45150	t
fixed-catch	fishing	2052	0.01518	0.01514	0.01522	per year
	mortality					
fixed-catch	total	2052	2997000	2989000	3004000	t
	biomass					
higher-F	biomass	2052	0.7427	0.7378	0.7519	ratio
	depletion					
higher-F	catch	2052	180200	179000	182400	t
higher-F	fishing	2052	0.08	0.08	0.08	per year
	mortality					
higher-F	total	2052	2343000	2328000	2372000	t
	biomass					
lower-F	biomass	2052	0.8678	0.8609	0.8719	ratio
	depletion					
lower-F	catch	2052	107400	106500	107900	t
lower-F	fishing	2052	0.04	0.04	0.04	per year
	mortality					
lower-F	total	2052	2738000	2716000	2751000	t
	biomass					

Table 7: Held-out prediction bias (predicted minus observed), RMSE and MASE by series and forecast lead. N counts scored forecasts; MASE uses each cutoff’s training naive-error scale.

Series	Lead	N	Bias	RMSE	MASE	Unit
Index 18 (Region 1)	1	5	-0.08847	0.10778	0.39718	relative
Index 18 (Region 1)	2	4	-0.095363	0.117	0.42491	relative
Index 18 (Region 1)	3	3	-0.093805	0.12075	0.41723	relative
Index 19 (Region 1)	1	5	-0.0691	0.18227	0.52469	relative
Index 19 (Region 1)	2	4	-0.0245	0.13014	0.41413	relative
Index 19 (Region 1)	3	3	-0.026698	0.1508	0.49854	relative
Index 20 (Region 2)	1	5	-0.2195	0.25108	0.84494	relative
Index 20 (Region 2)	2	4	-0.26202	0.28574	0.99863	relative

Series	Lead	N	Bias	RMSE	MASE	Unit
Index 20 (Region 2)	3	3	-0.30353	0.32467	1.1481	relative

15 Figures

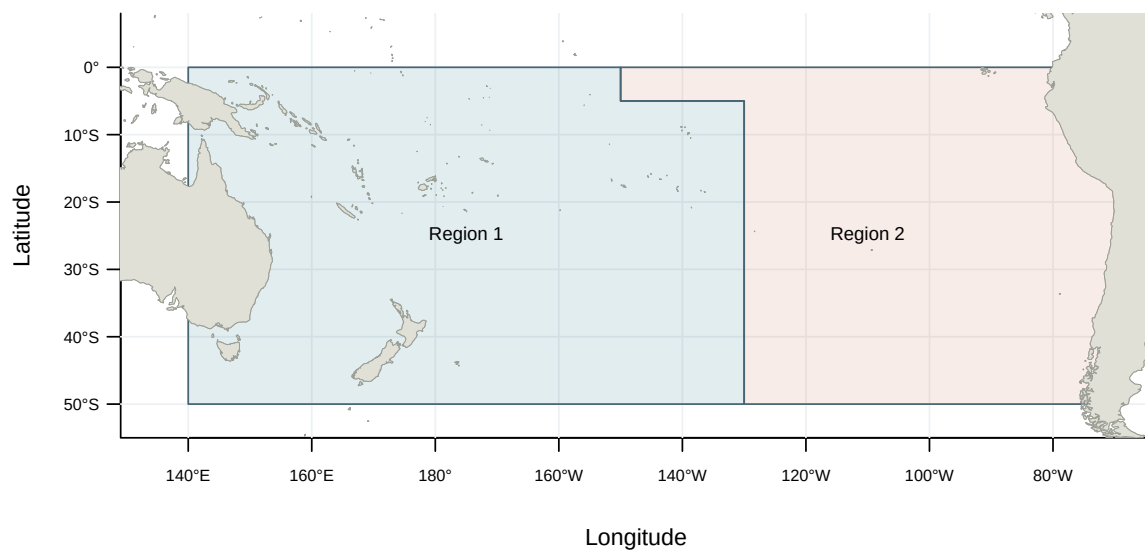


Figure 1: Model regions for the 2024 South Pacific albacore assessment (WCPFC-SC20-SA-WP-02, Rev.03, Figure 2).



Figure 2: Illustrative output. Catch by fishing group.

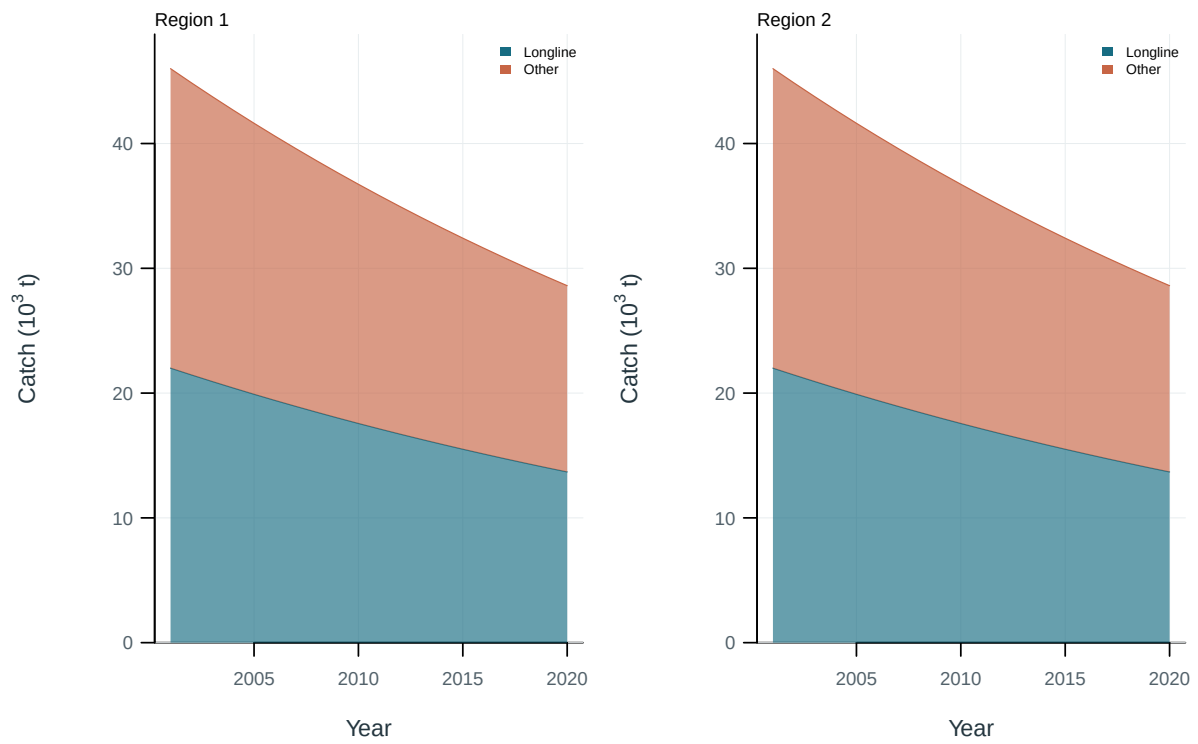


Figure 3: Illustrative output. Catch by fishing group and region.

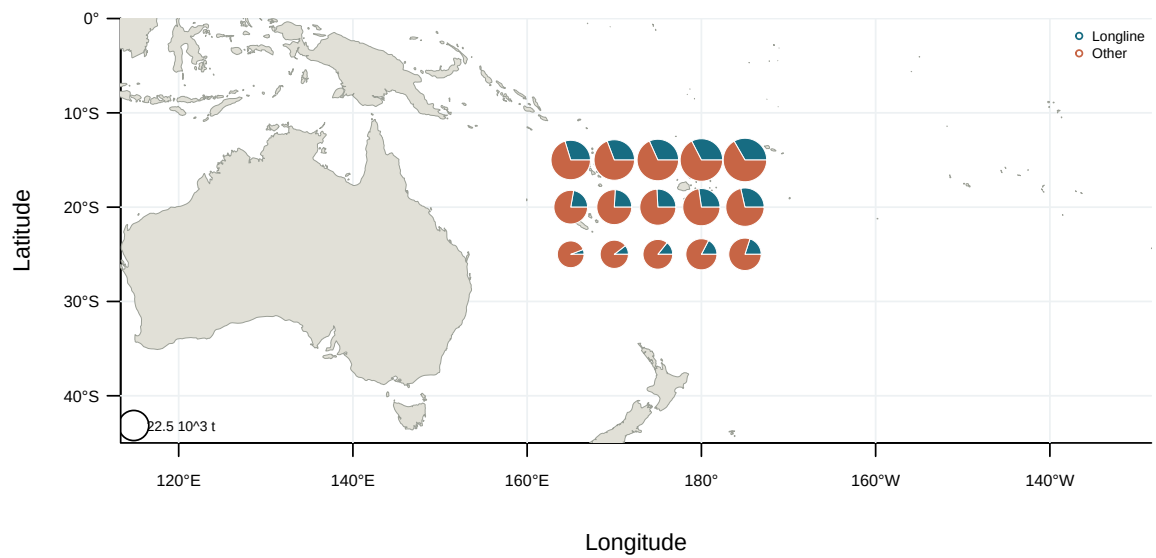


Figure 4: Illustrative output. Spatial catch distribution by fishing group.



Figure 5: Observation availability by data type and fishery. Coloured cells indicate years with observations.

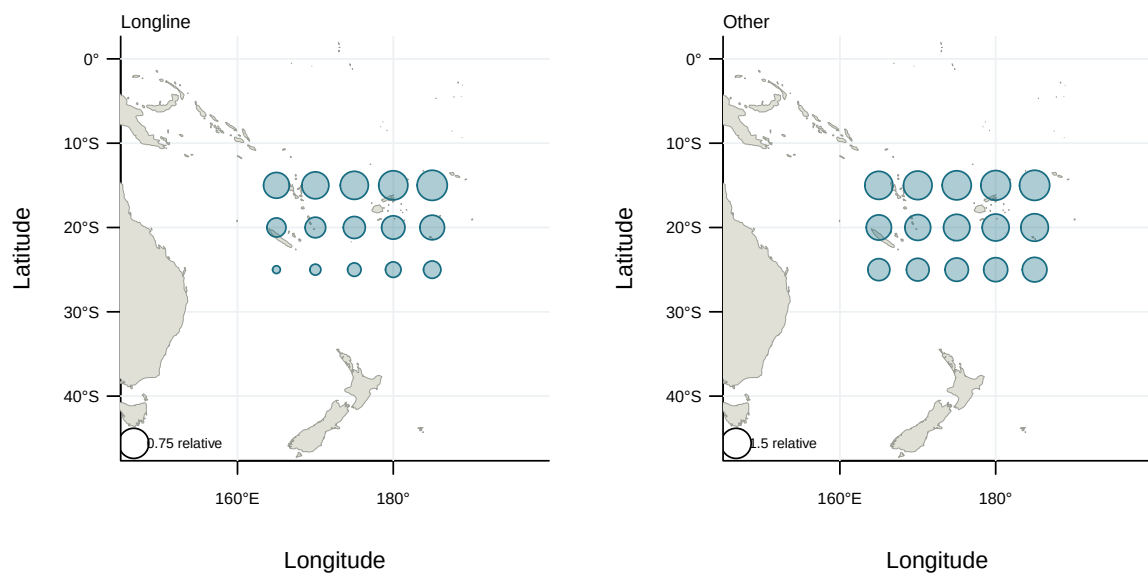


Figure 6: Illustrative output. Spatial distribution of nominal abundance indices.

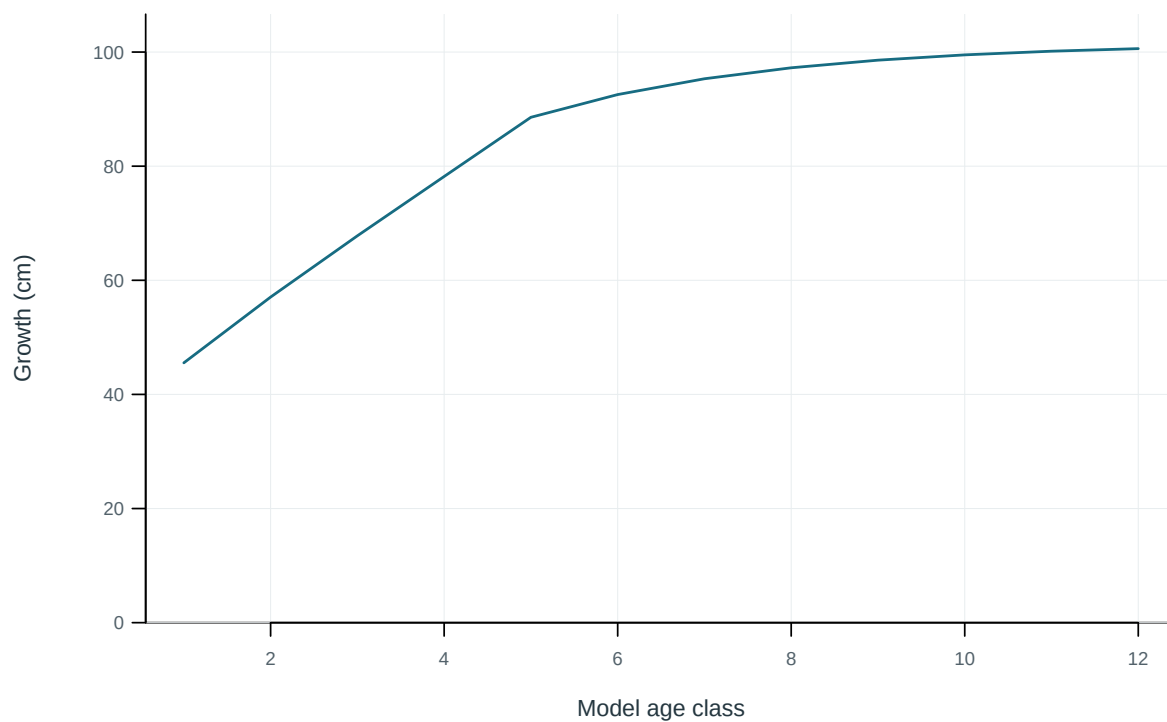


Figure 7: Mean length at the start of each model age class.

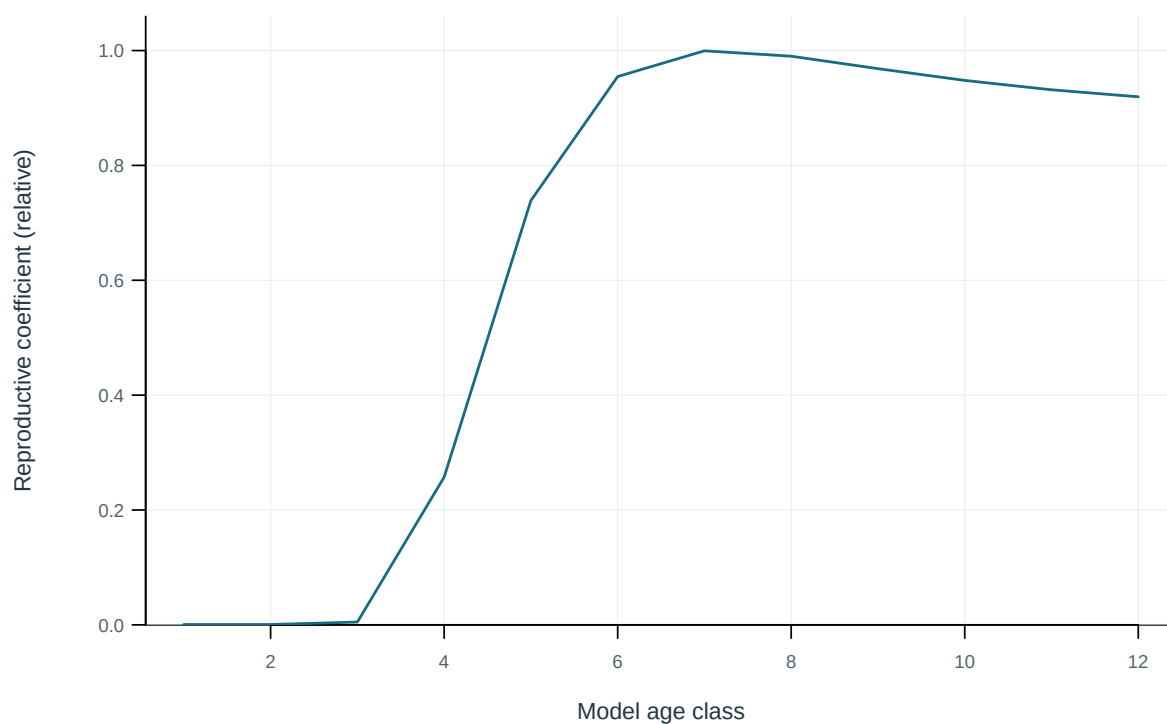


Figure 8: Model reproductive coefficient at age.

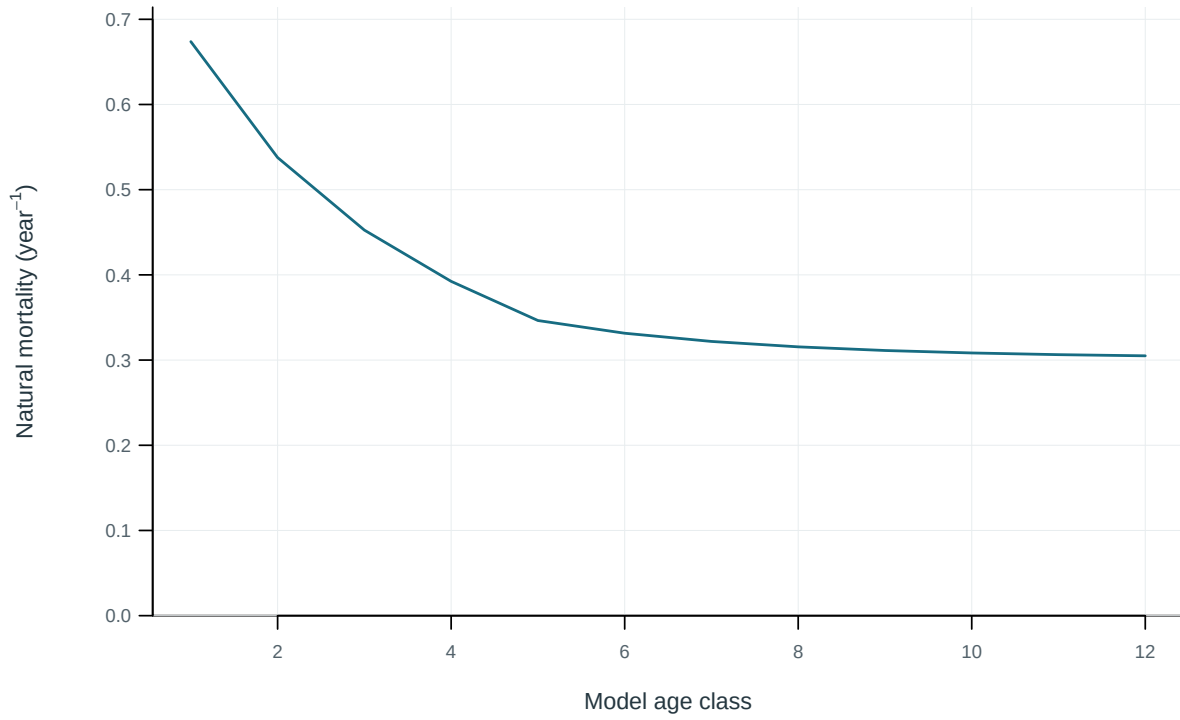


Figure 9: Annual instantaneous natural mortality at age.

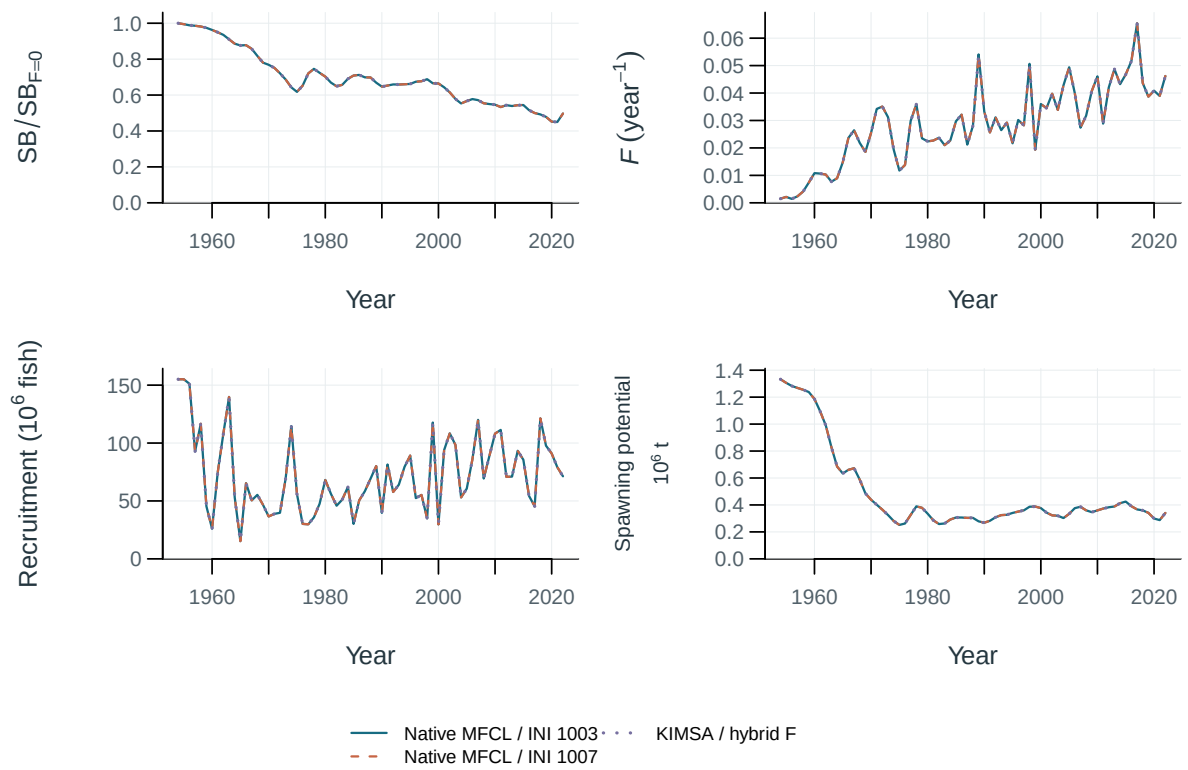


Figure 10: Stepwise model trajectories.

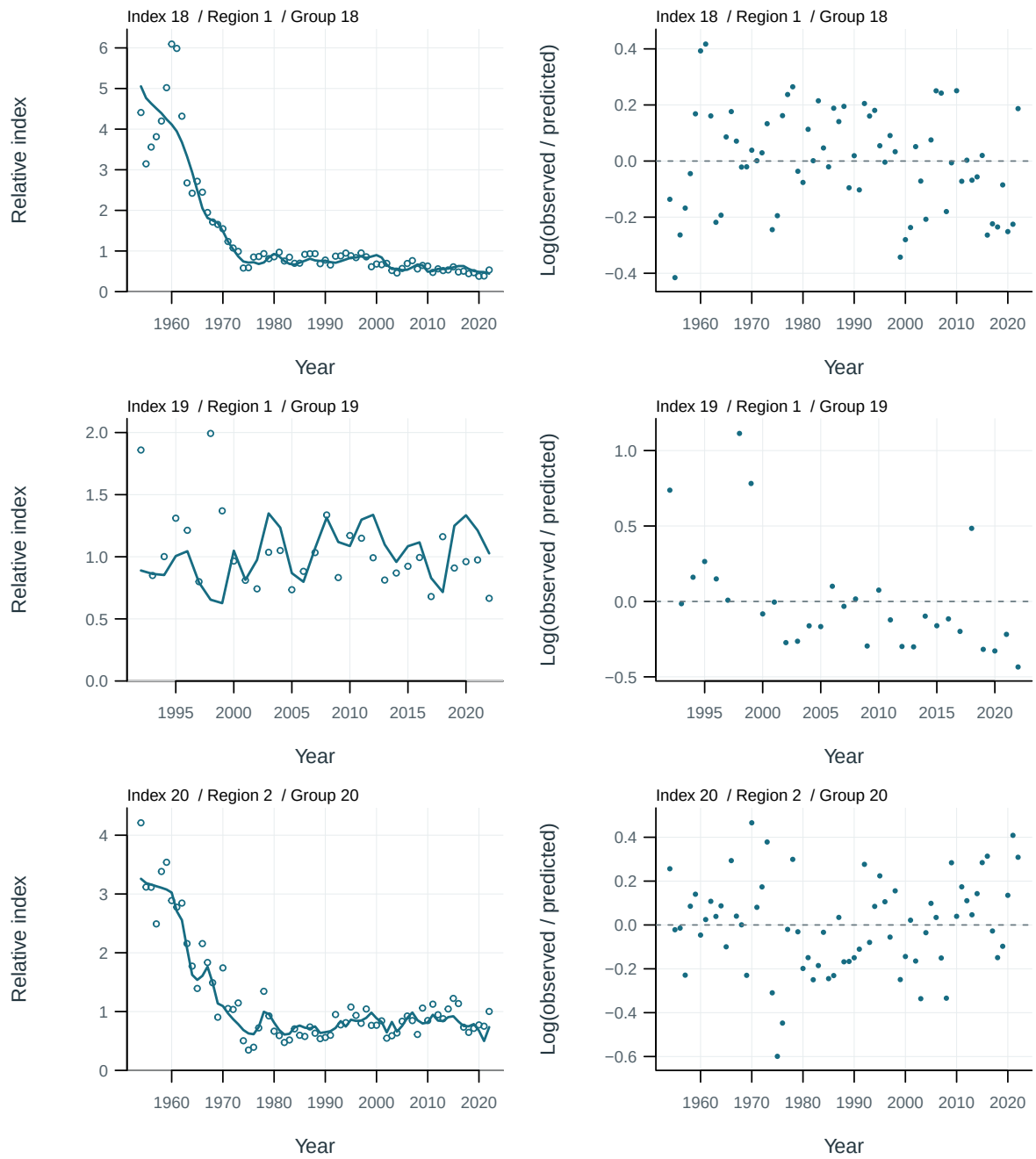


Figure 11: Abundance-index fits and log residuals by series. Residuals are $\log(\text{observed}/\text{predicted})$.

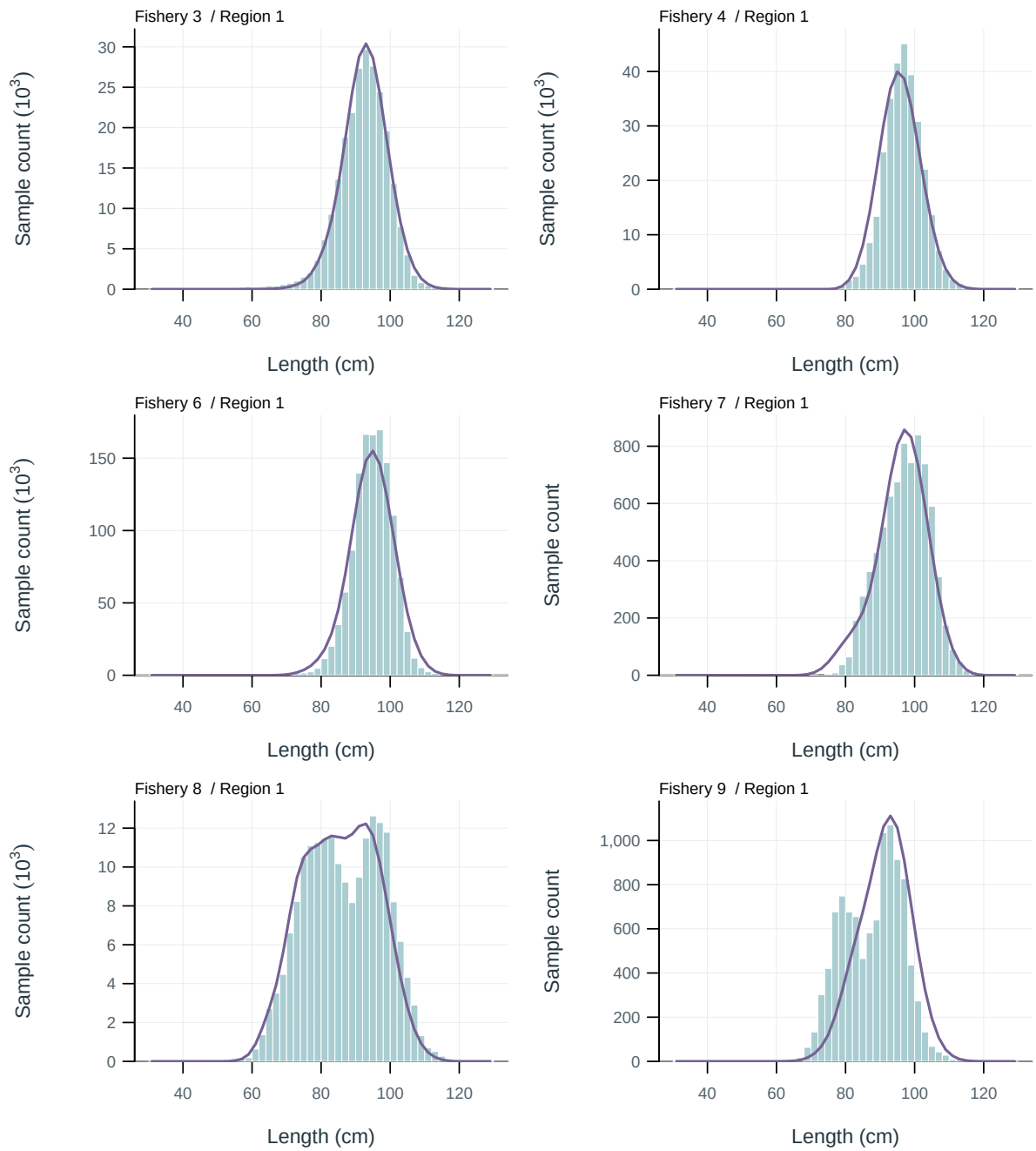


Figure 12: Length-composition fits by fishery. Observed sample counts (bars) and predictions (lines), pooled over years.

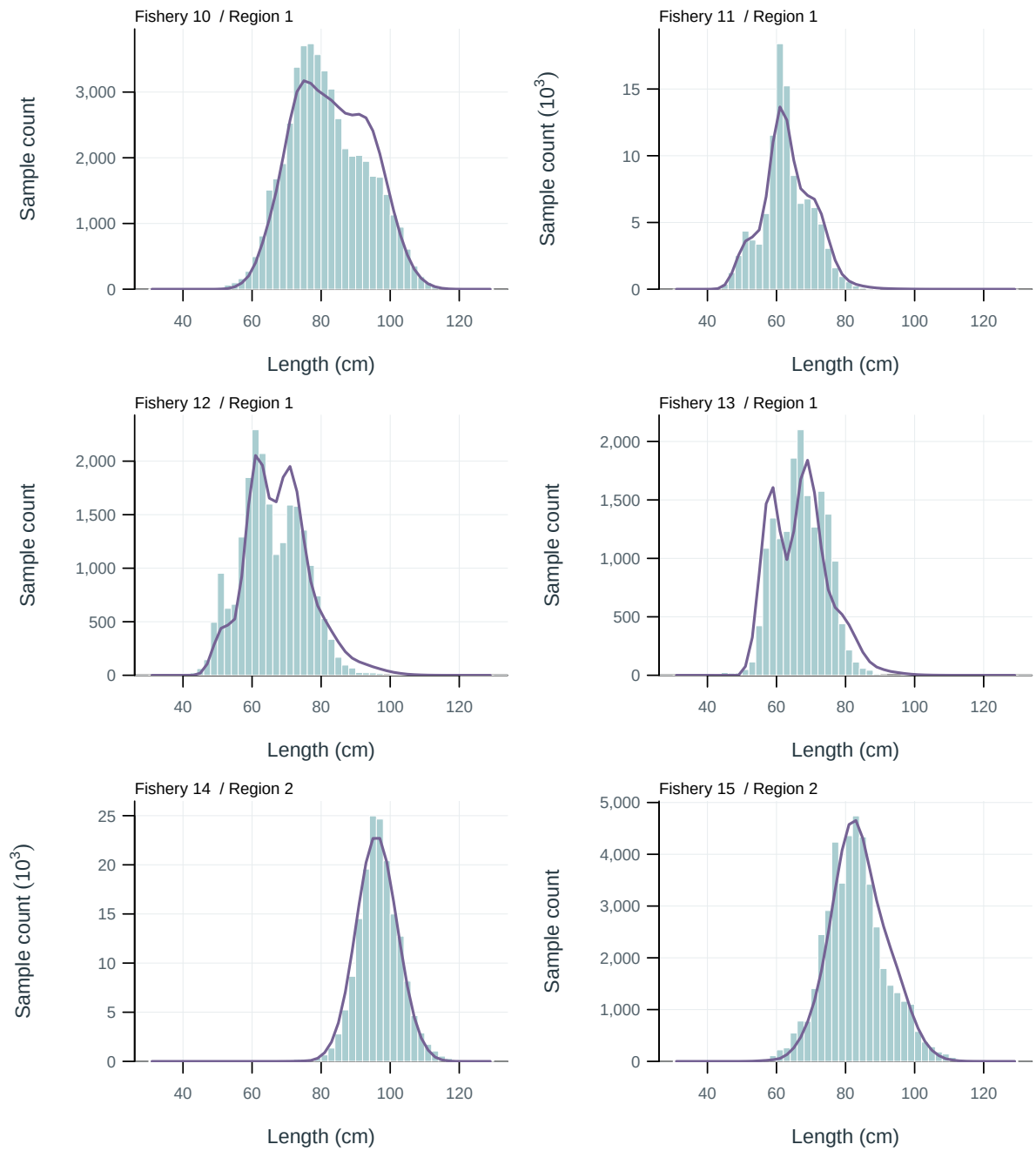


Figure 13: Length-composition fits by fishery. Observed sample counts (bars) and predictions (lines), pooled over years.

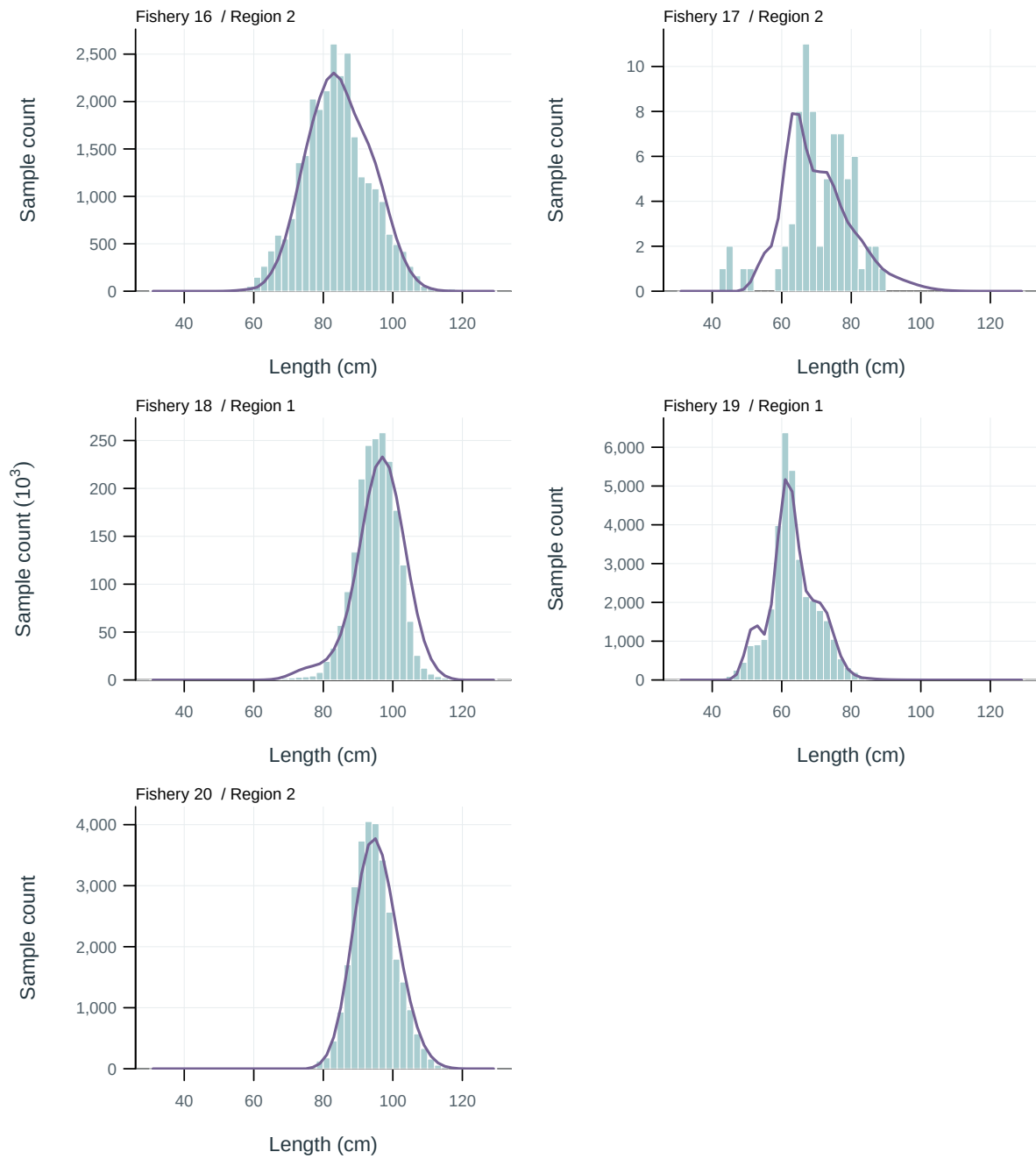


Figure 14: Length-composition fits by fishery. Observed sample counts (bars) and predictions (lines), pooled over years.

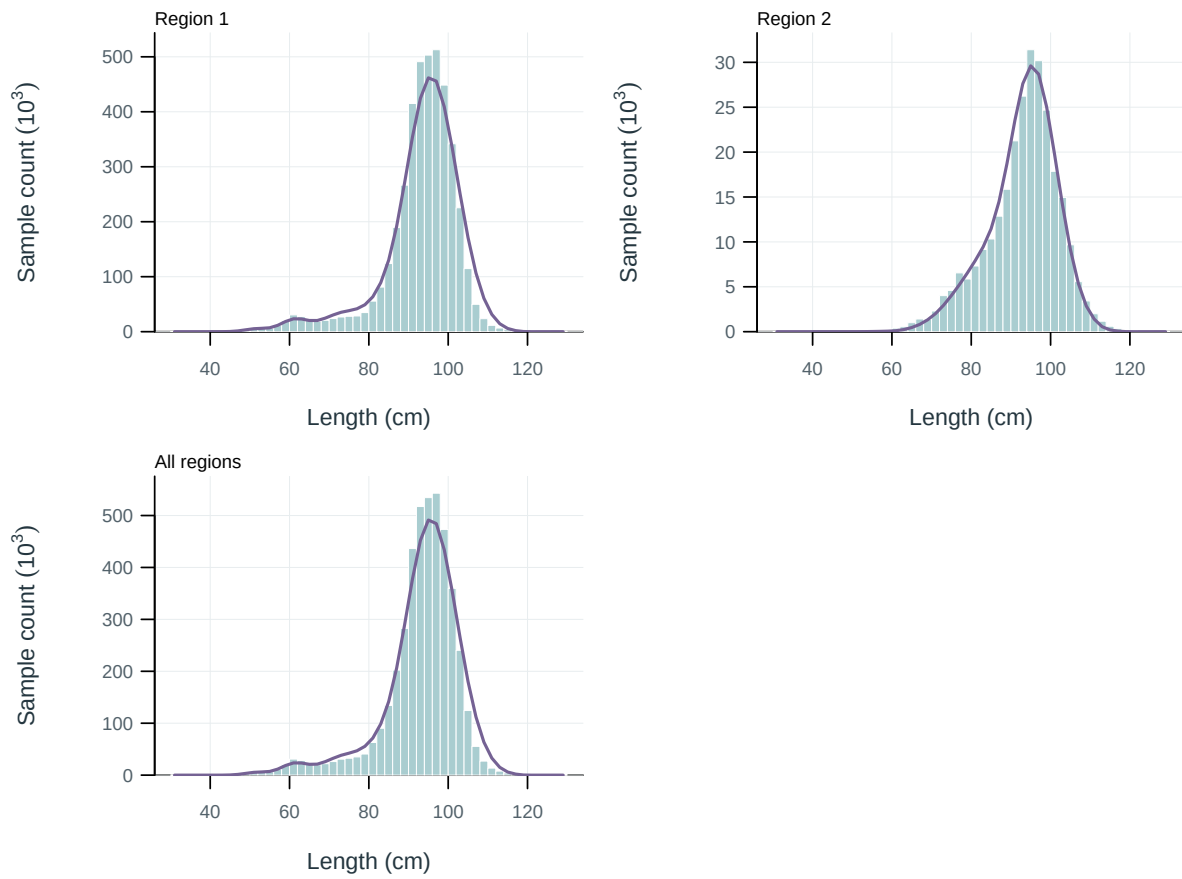


Figure 15: Length-composition fits by region. Observed sample counts (bars) and predictions (lines), pooled over years.

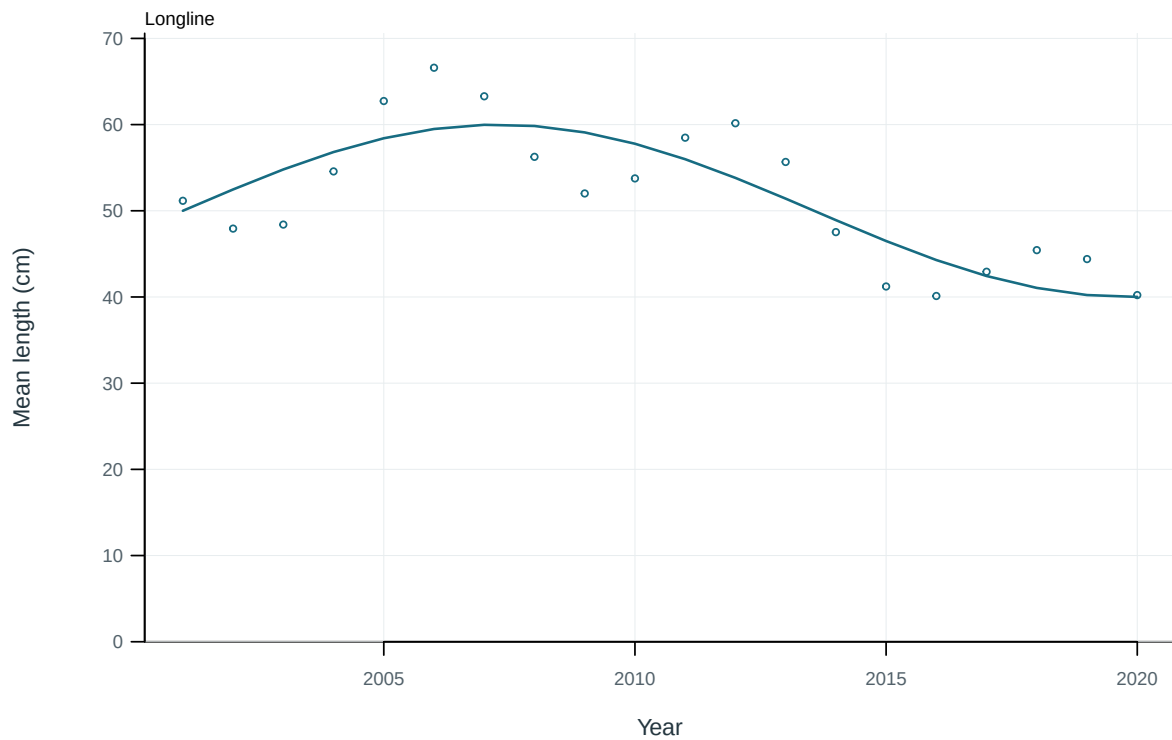


Figure 16: Illustrative output. Observed and predicted length summaries for longline fisheries.



Figure 17: Illustrative output. Observed and predicted length summaries for other fisheries.

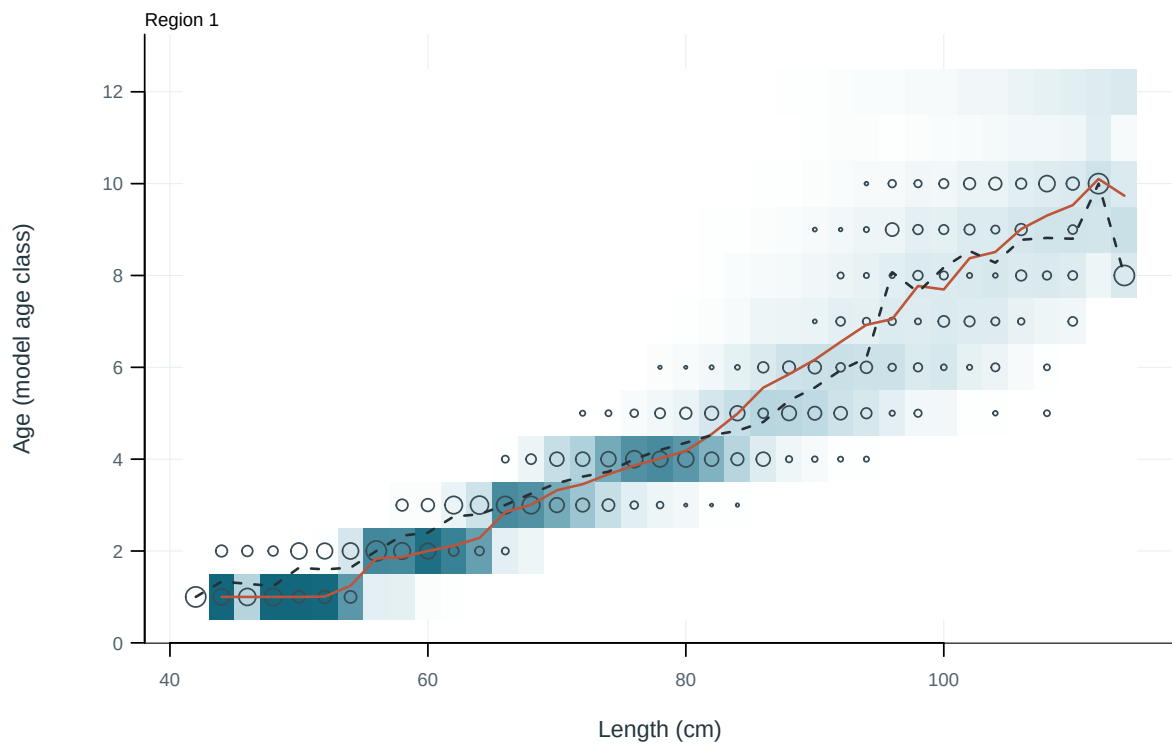


Figure 18: Observed and predicted age distributions conditional on length. Cells: predictions; circles: observations; solid and dashed lines: predicted and observed mean age.

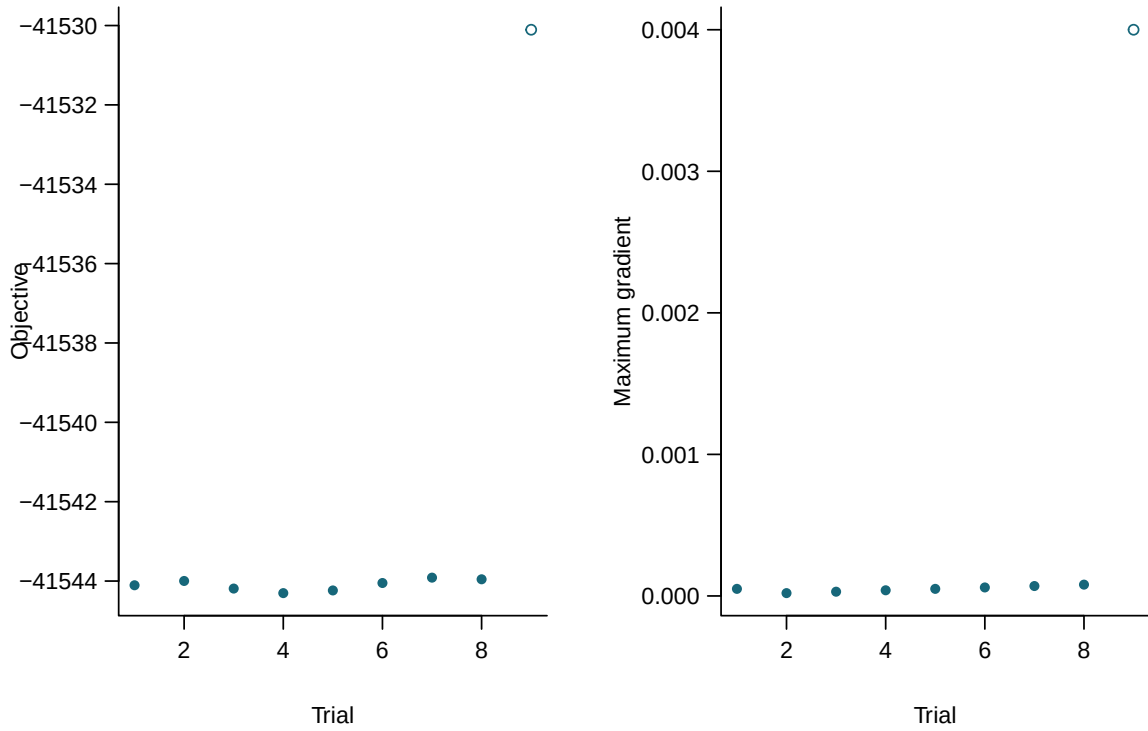


Figure 19: Final objectives and maximum gradients for independent starting values. Filled points meet all recorded numerical checks.

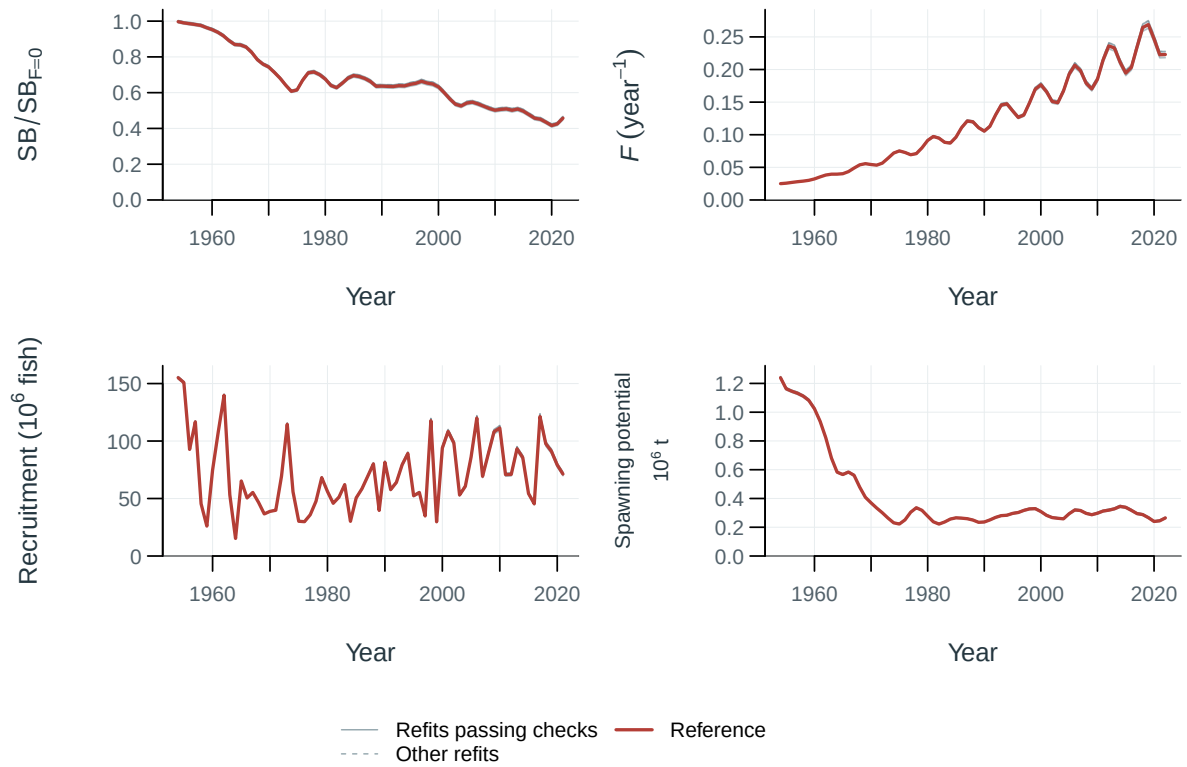


Figure 20: Jitter model trajectories.

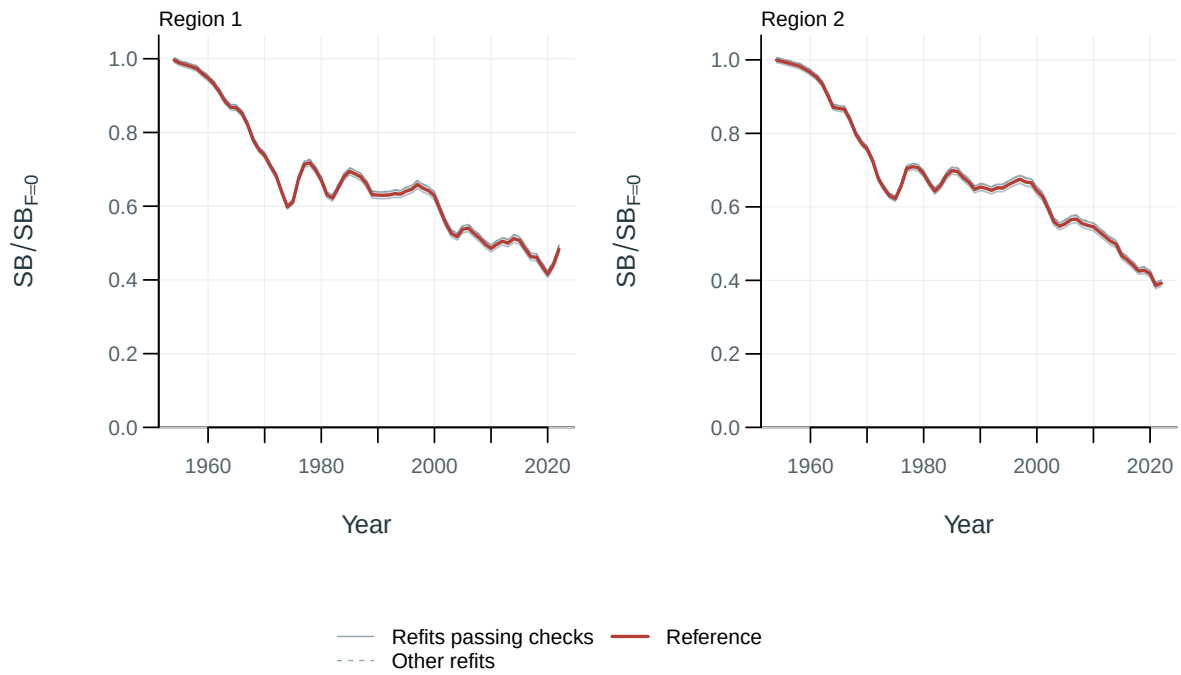


Figure 21: Spawning depletion by model and region, relative to each model's dynamic unfished reference.

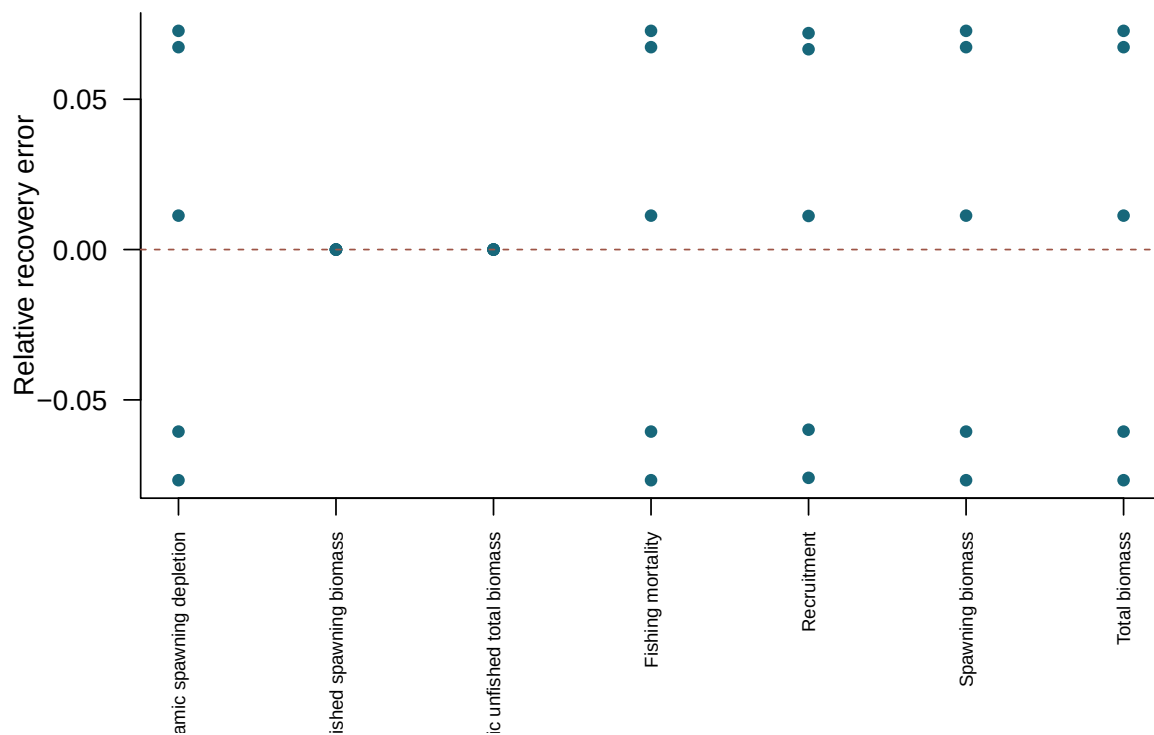


Figure 22: Relative recovery error in terminal quantities for converged simulated-data refits.

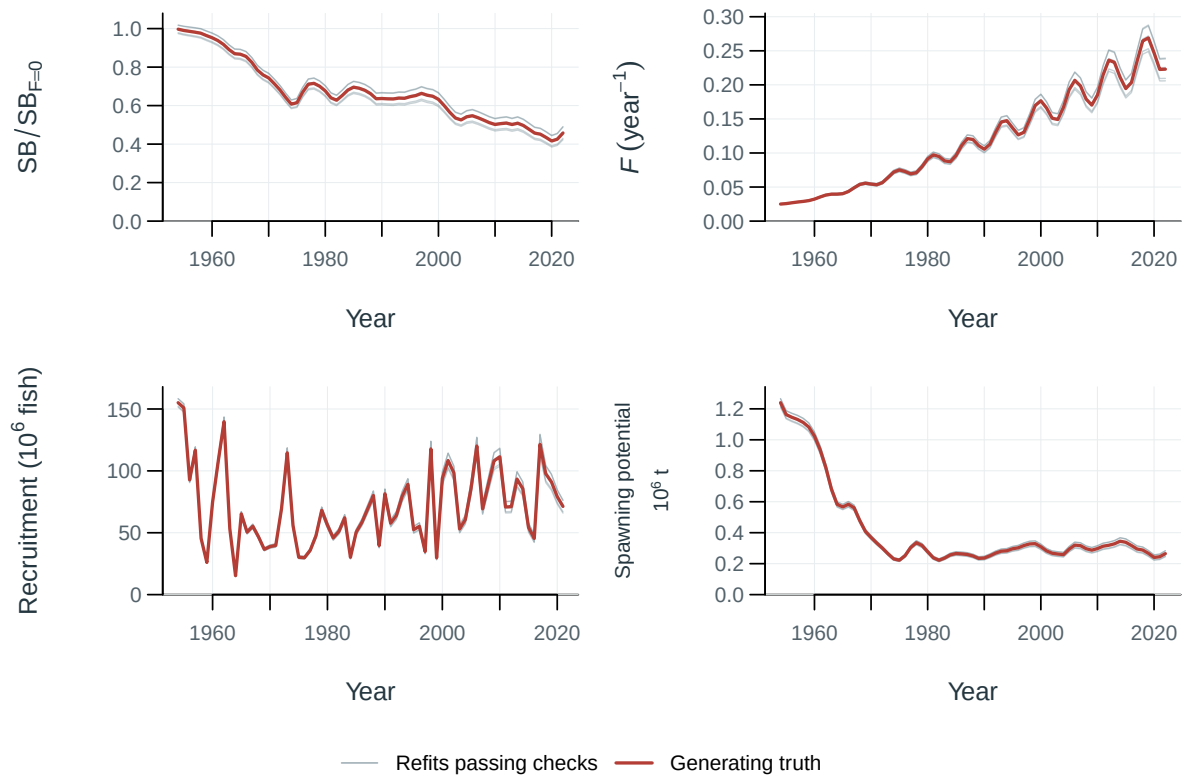


Figure 23: Selftest model trajectories.

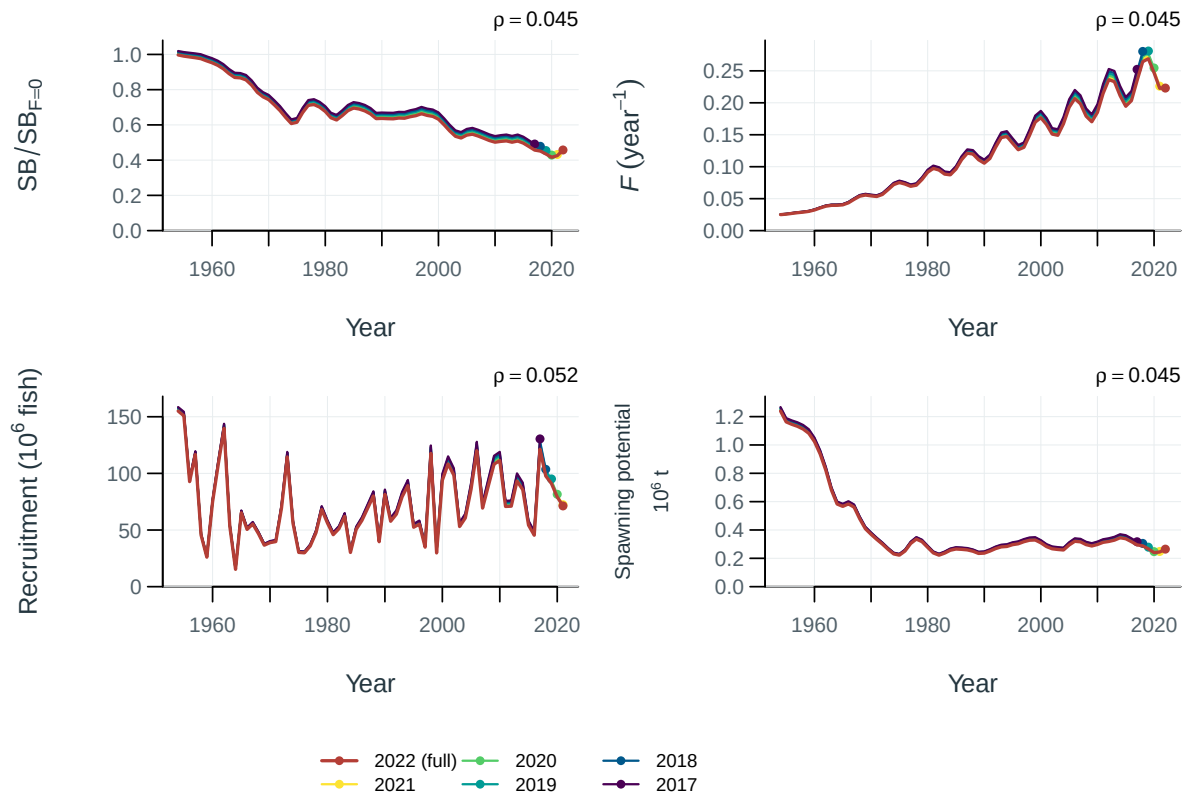


Figure 24: Population trajectories for the full-data fit and retrospective peels. Points mark terminal years; rho is Mohn's mean terminal relative difference for peels passing the recorded numerical checks.

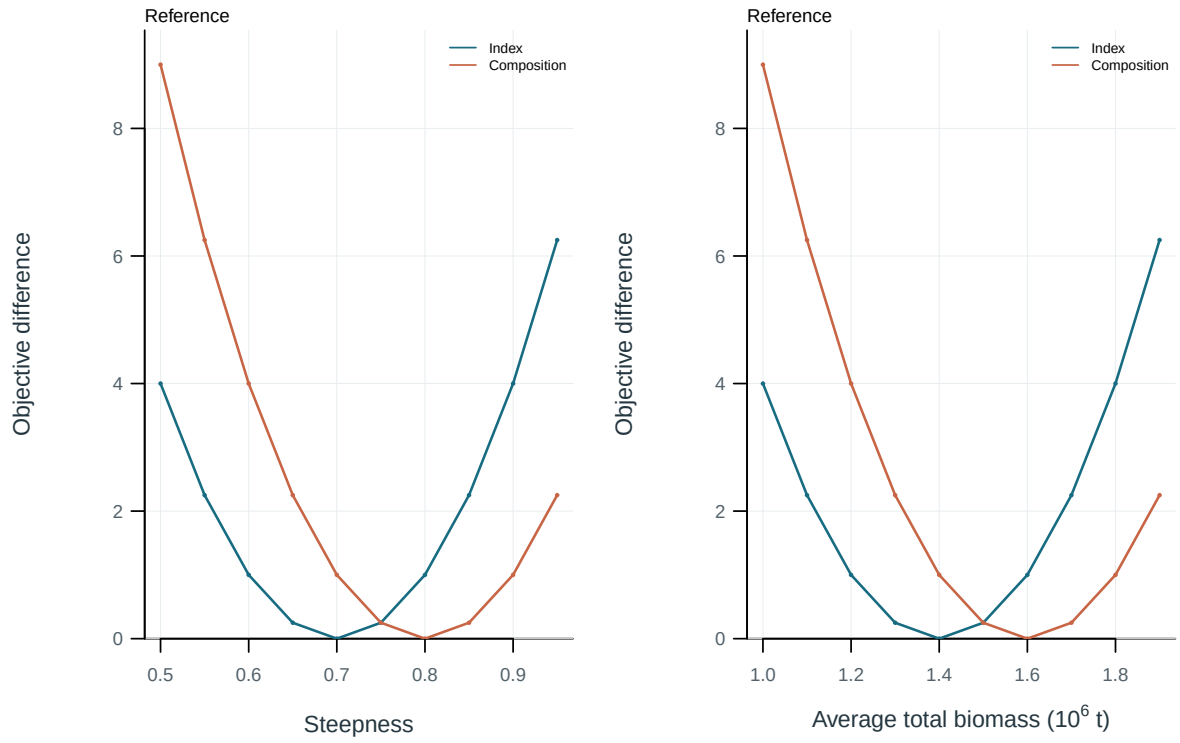


Figure 25: Total objective and component profiles. Each curve is centered on its own minimum. Profile on the supplied parameter scale.; Total biomass averaged over the declared model periods, 2001-2020.

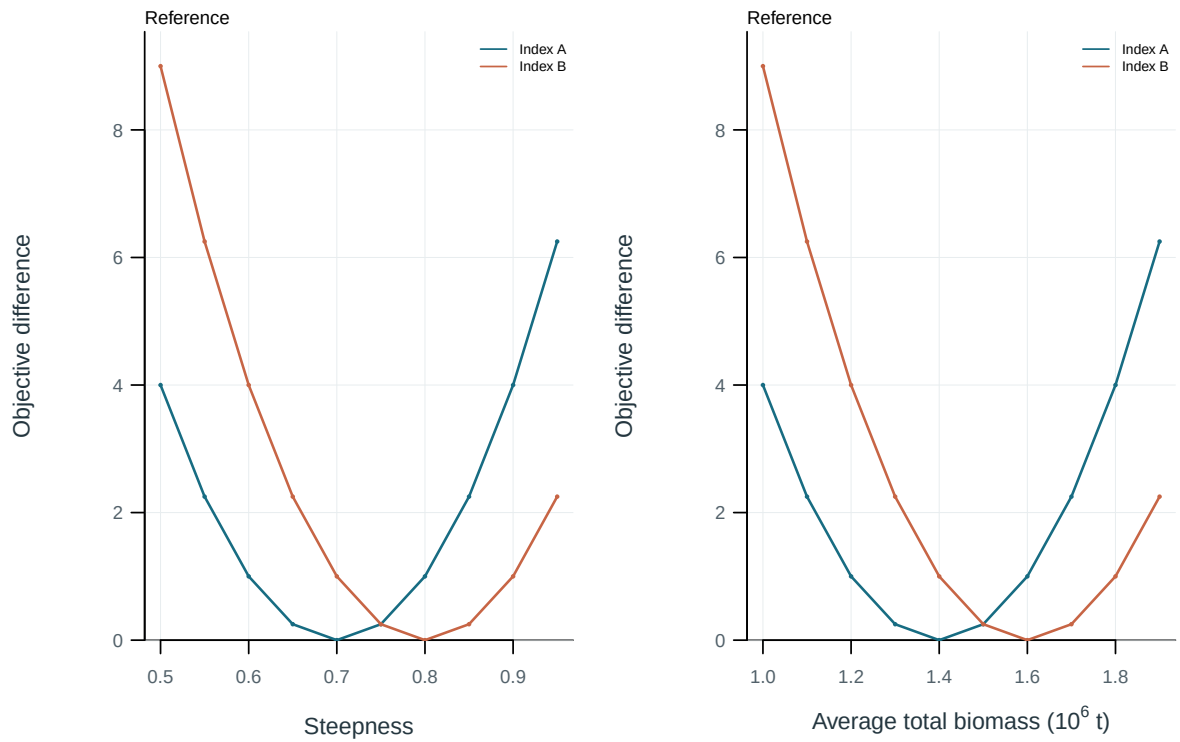


Figure 26: Objective profiles for abundance-index components. Each curve is centered on its own minimum. Profile on the supplied parameter scale.; Total biomass averaged over the declared model periods, 2001-2020.

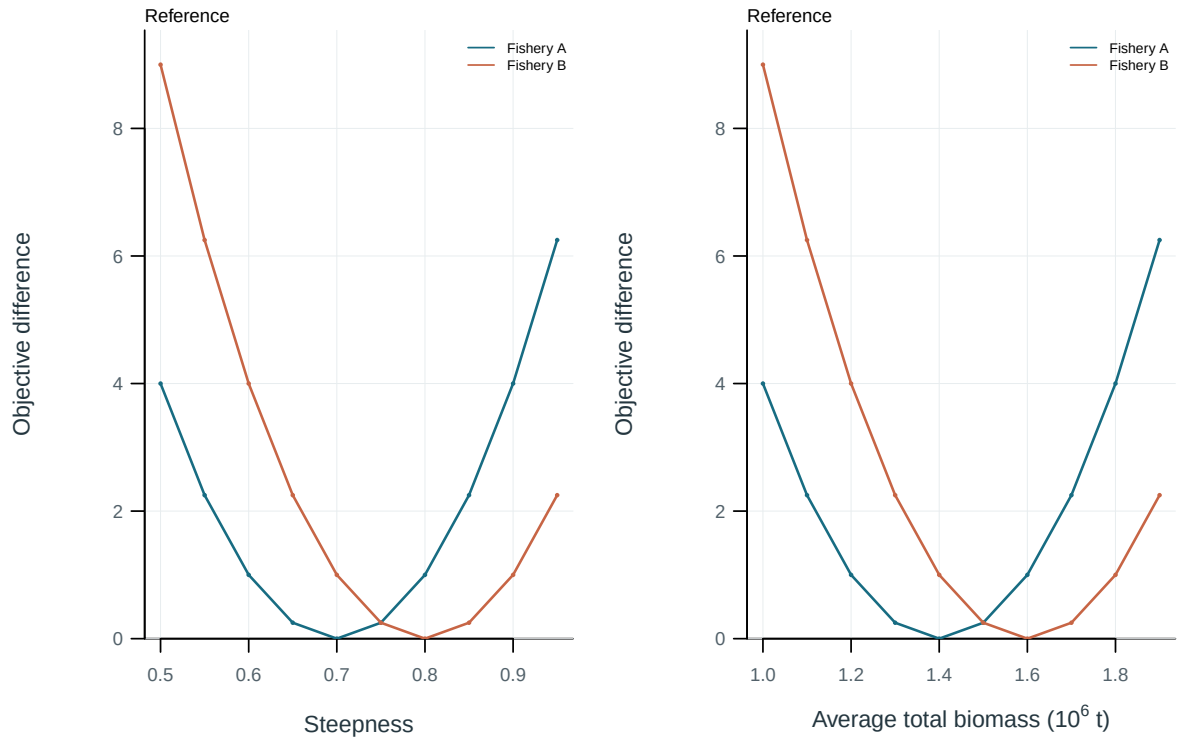


Figure 27: Objective profiles for length-composition components. Each curve is centered on its own minimum. Profile on the supplied parameter scale.; Total biomass averaged over the declared model periods, 2001-2020.

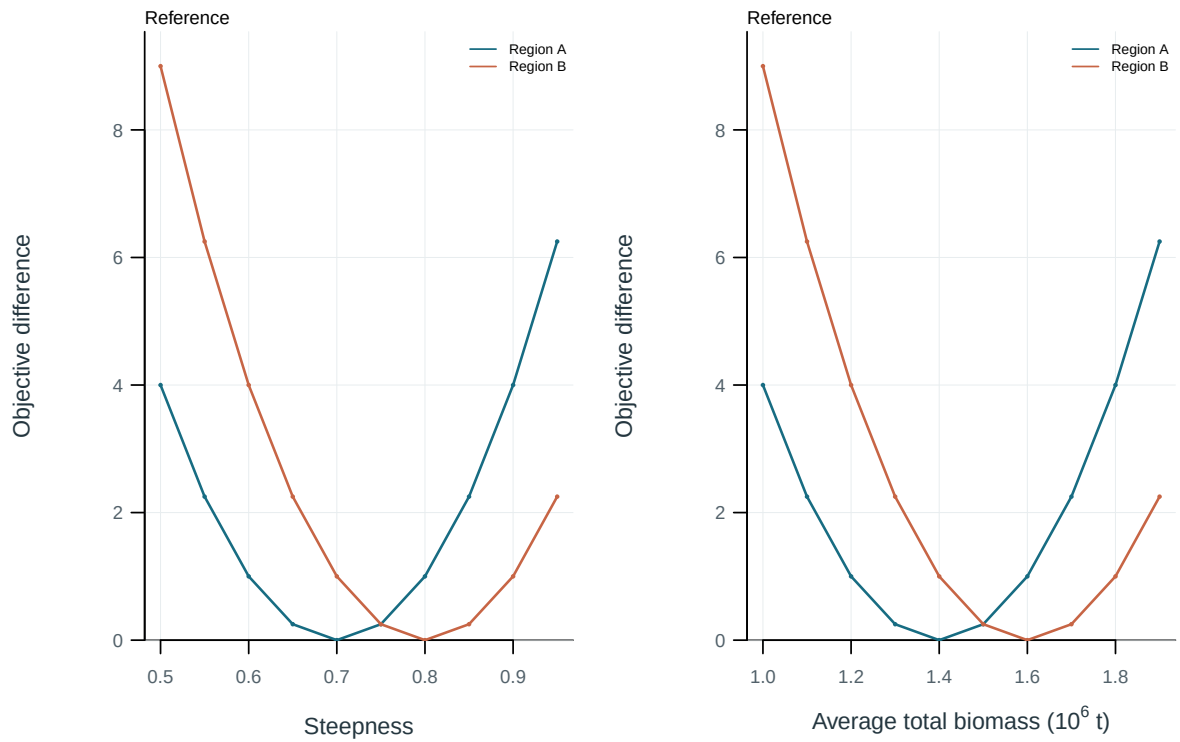


Figure 28: Objective profiles for conditional age-at-length components. Each curve is centered on its own minimum. Profile on the supplied parameter scale.; Total biomass averaged over the declared model periods, 2001-2020.

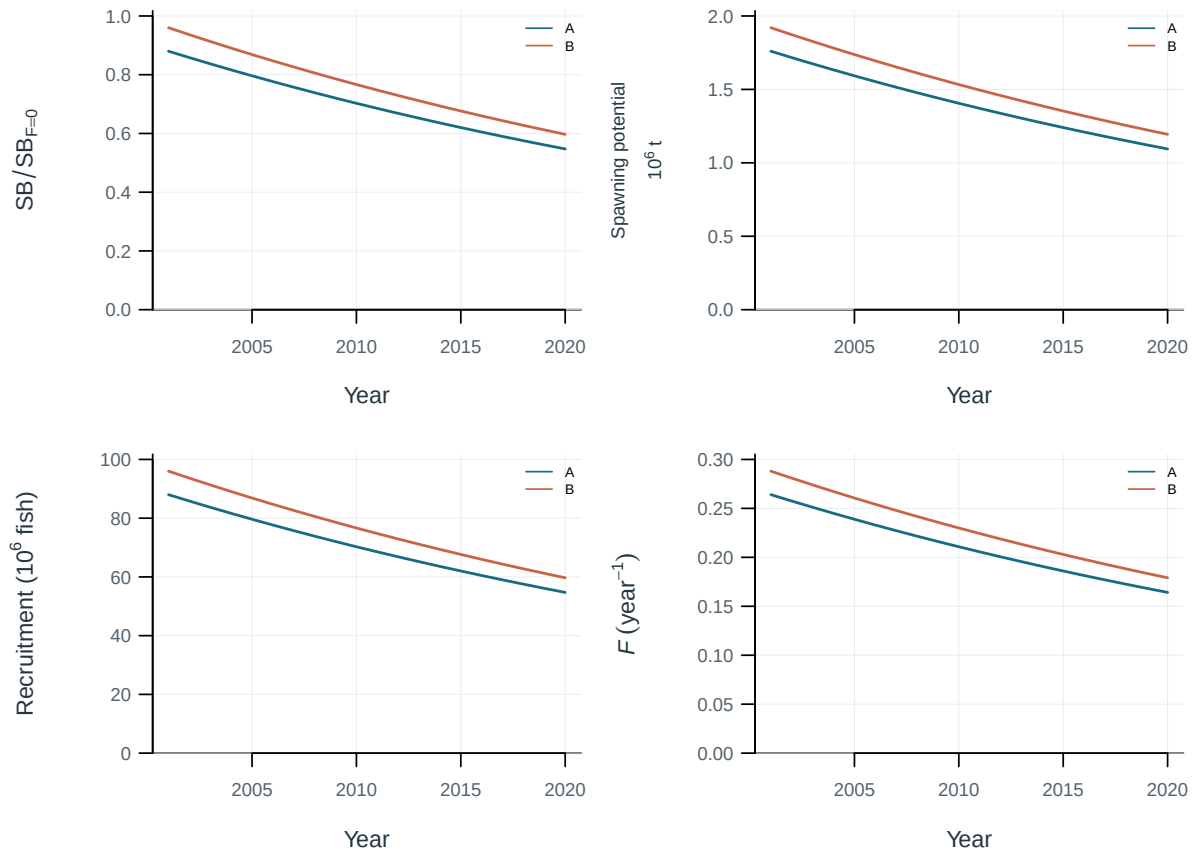


Figure 29: Illustrative output. Diagnostic and production-model trajectories.

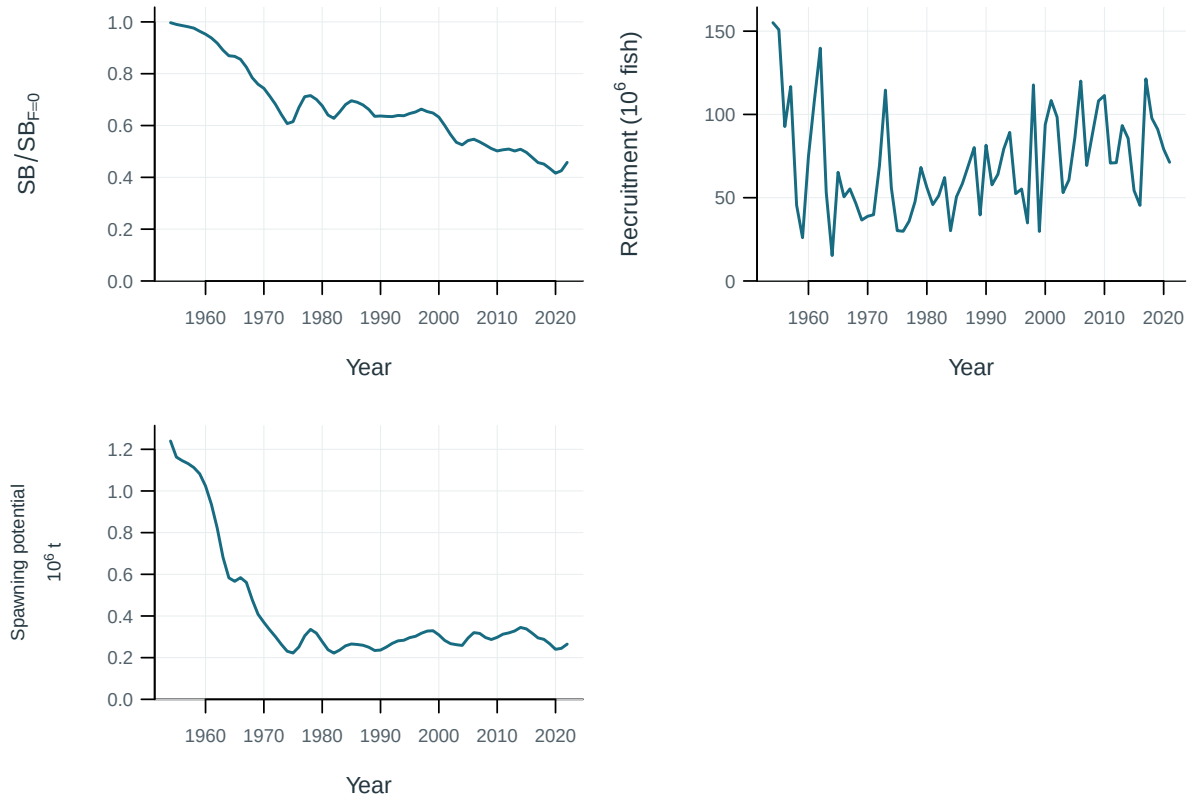


Figure 30: Estimated population trajectories.

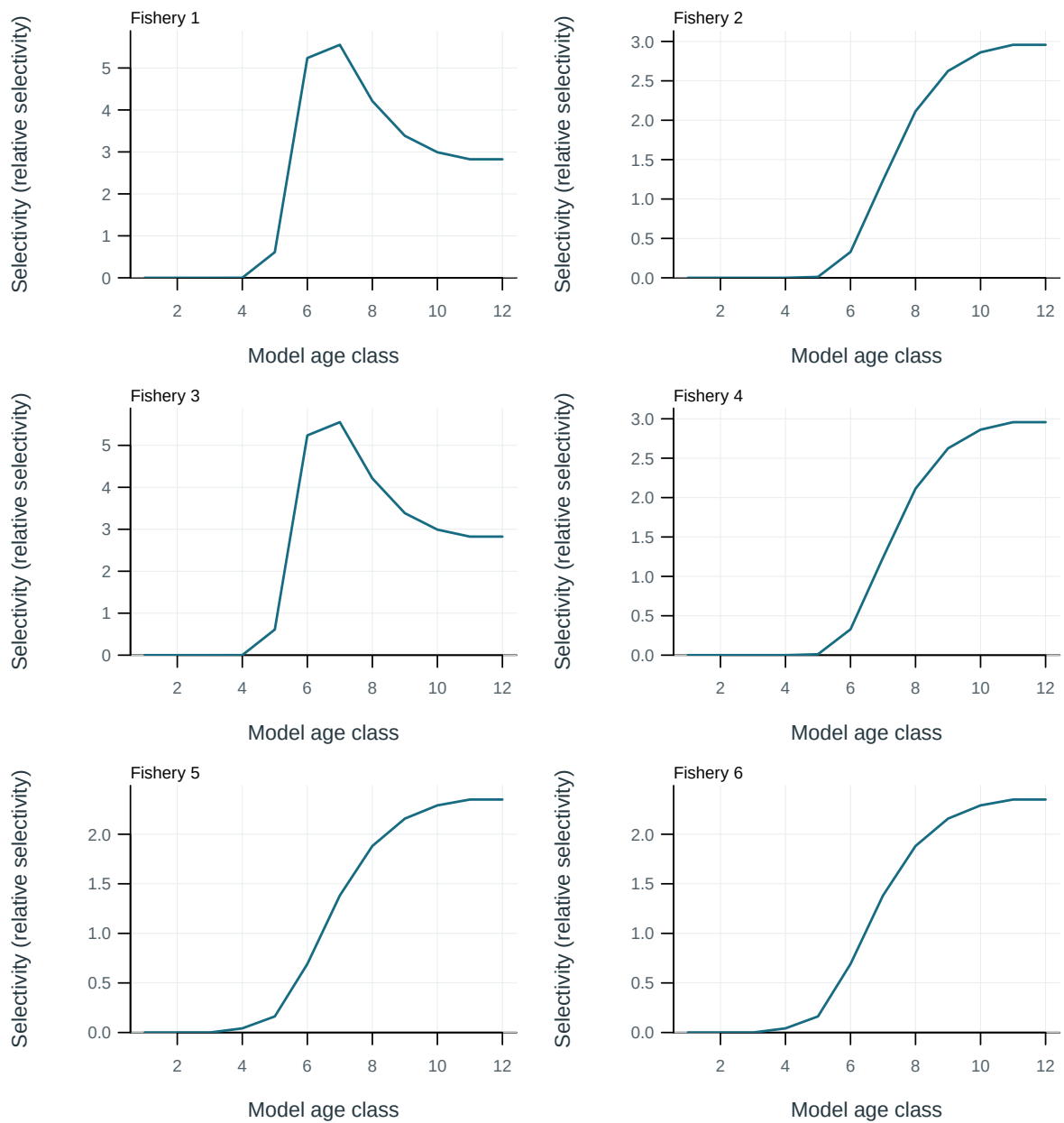


Figure 31: Fishery selectivity at age.

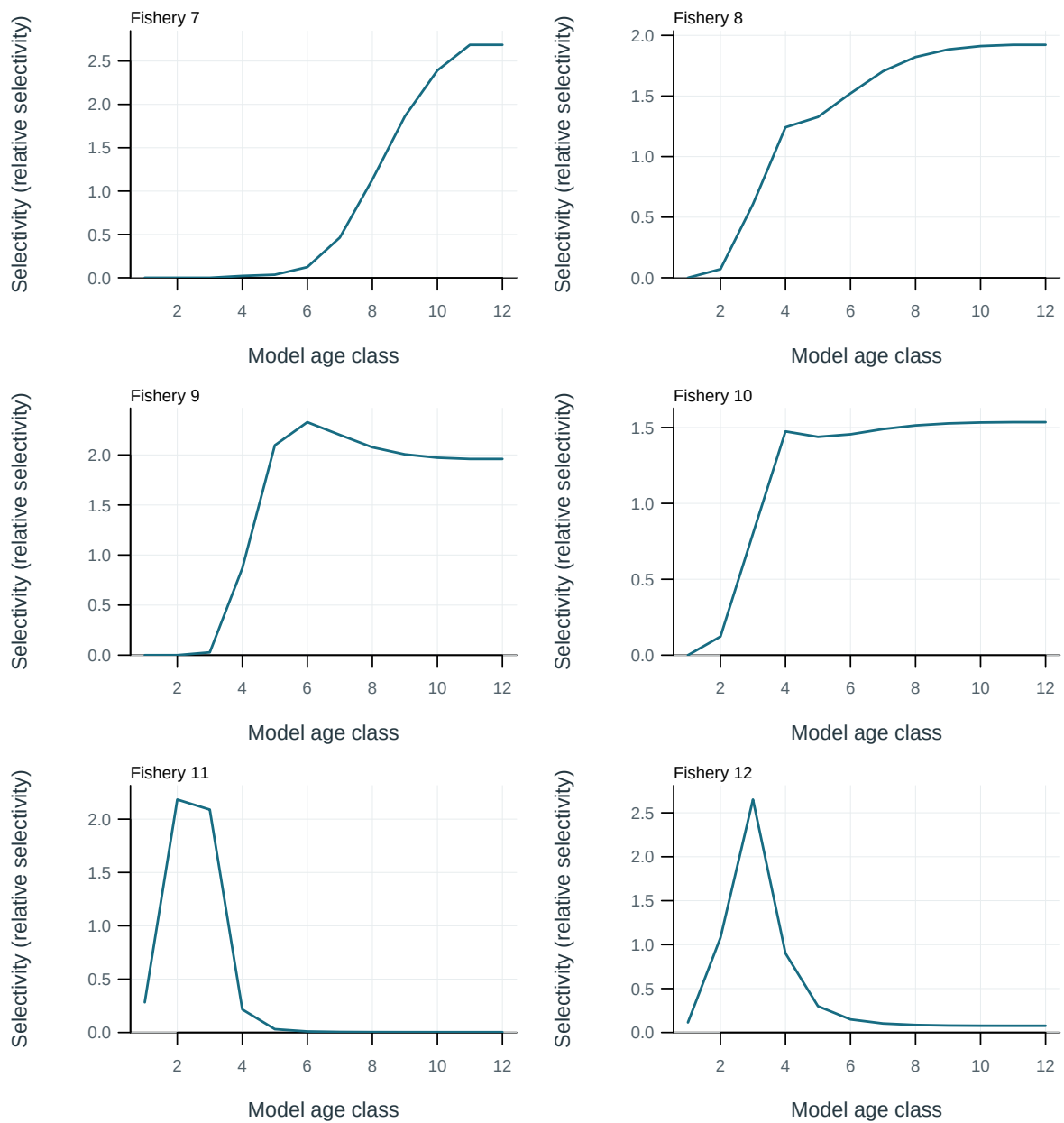


Figure 32: Fishery selectivity at age.

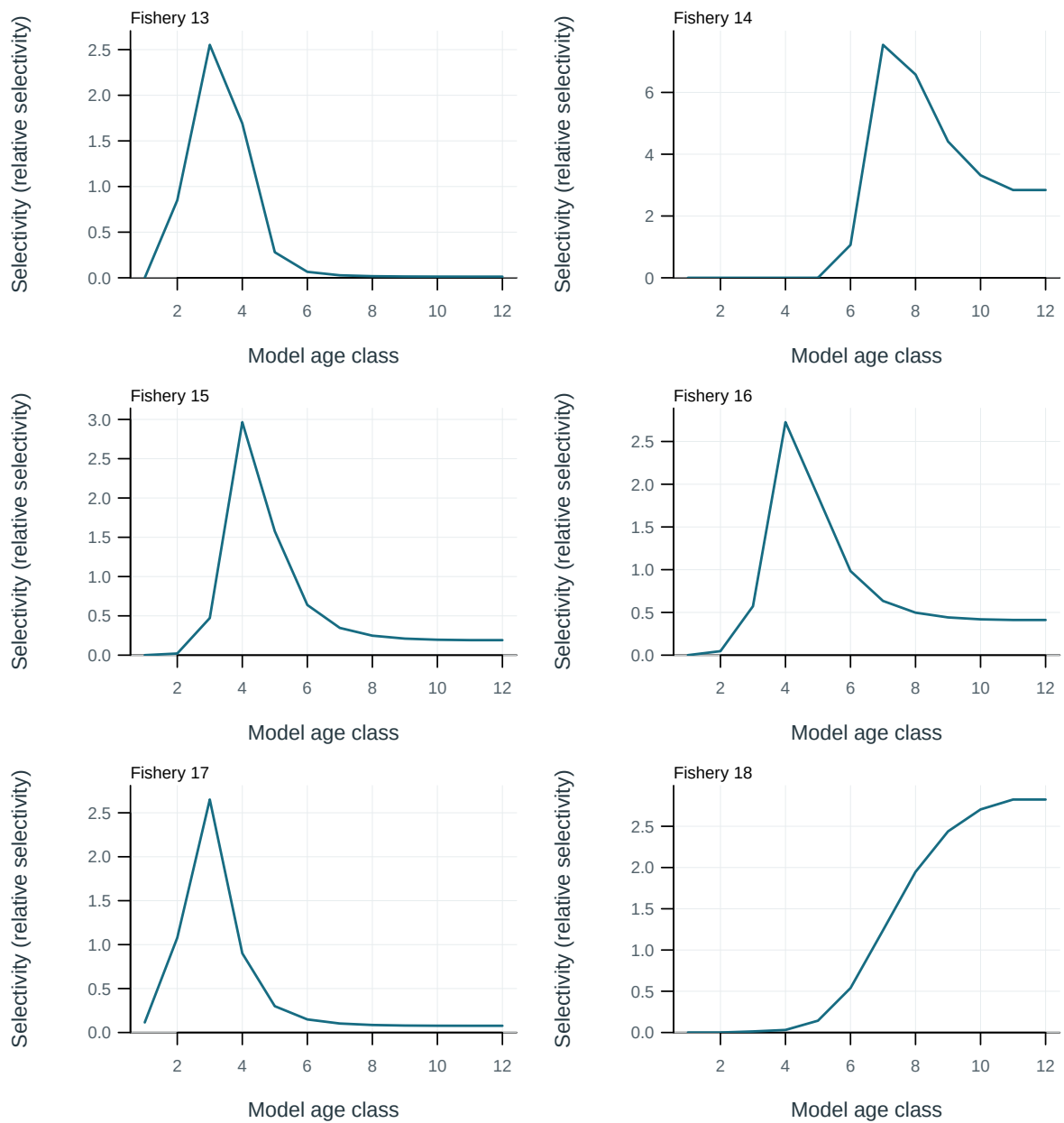


Figure 33: Fishery selectivity at age.

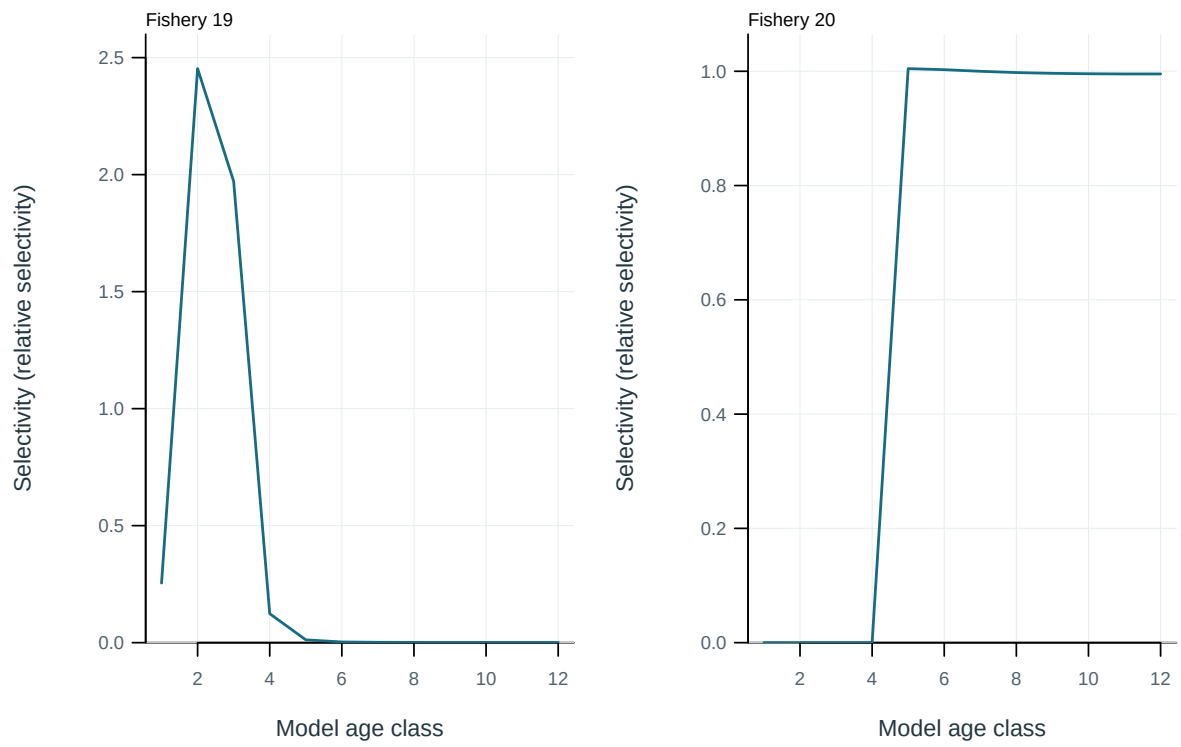


Figure 34: Fishery selectivity at age.

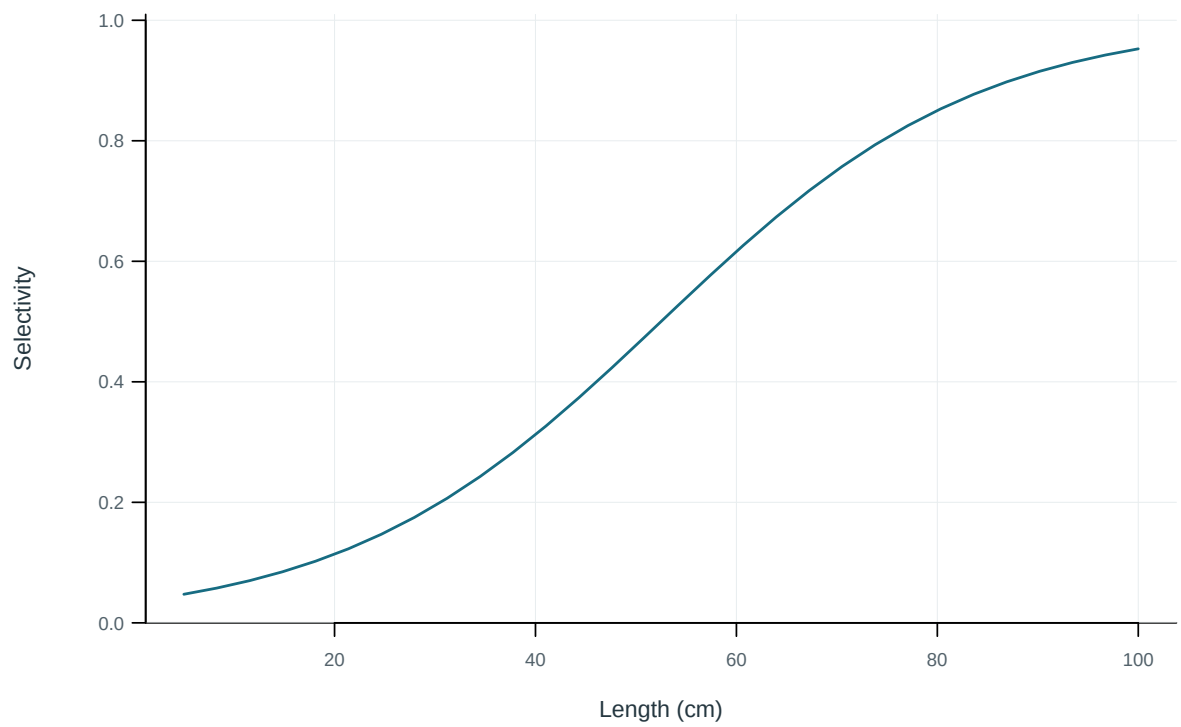


Figure 35: Illustrative output. Selectivity at length by fishery.

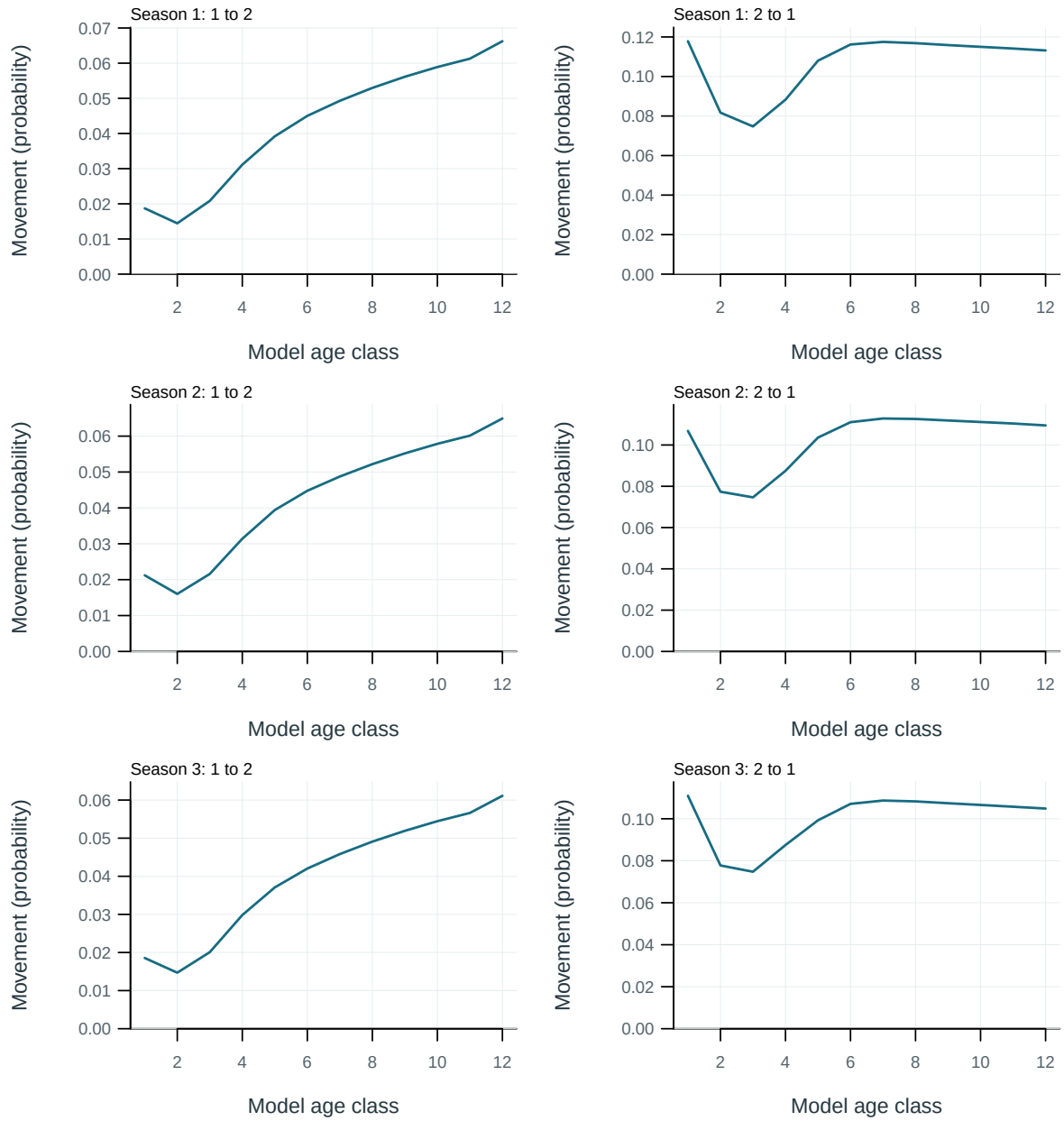


Figure 36: Probability of movement from source to destination region per movement event.

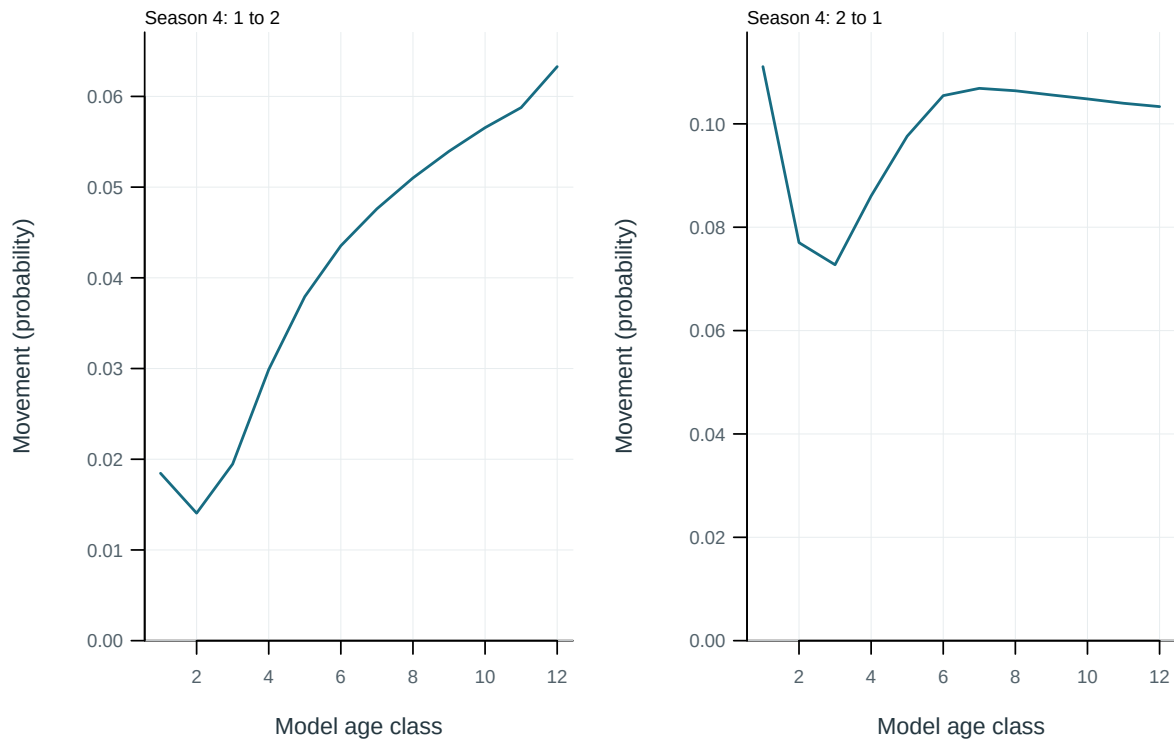


Figure 37: Probability of movement from source to destination region per movement event.

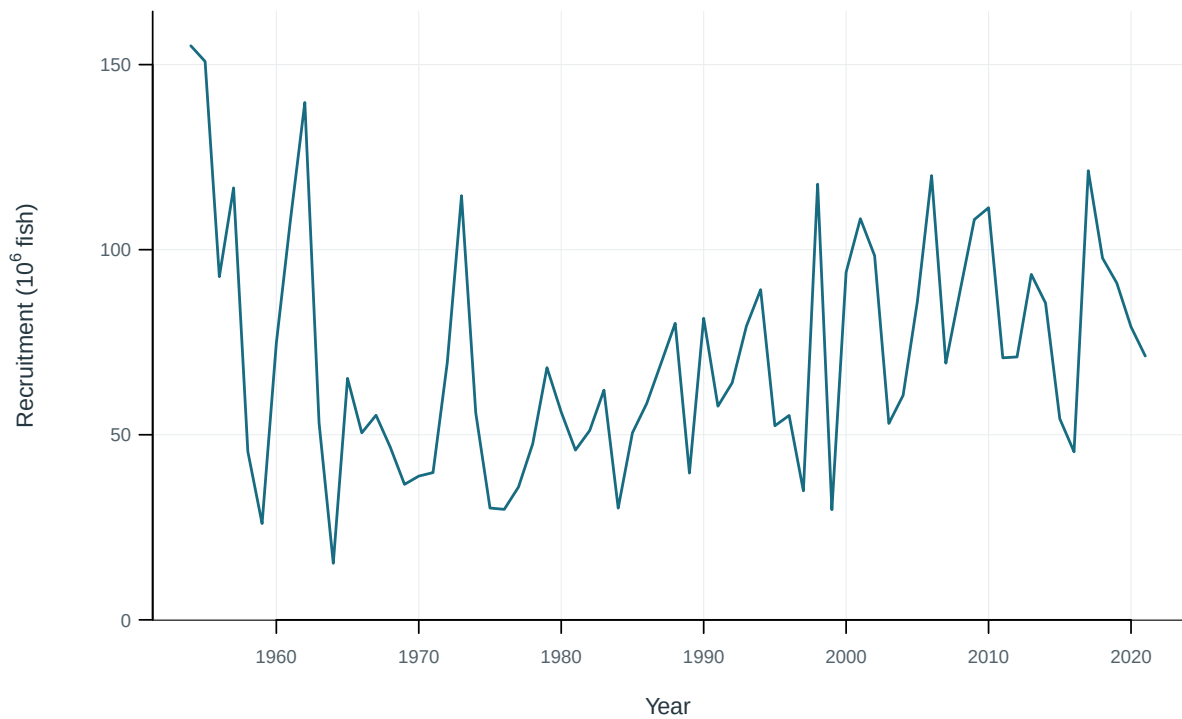


Figure 38: Annual recruitment.

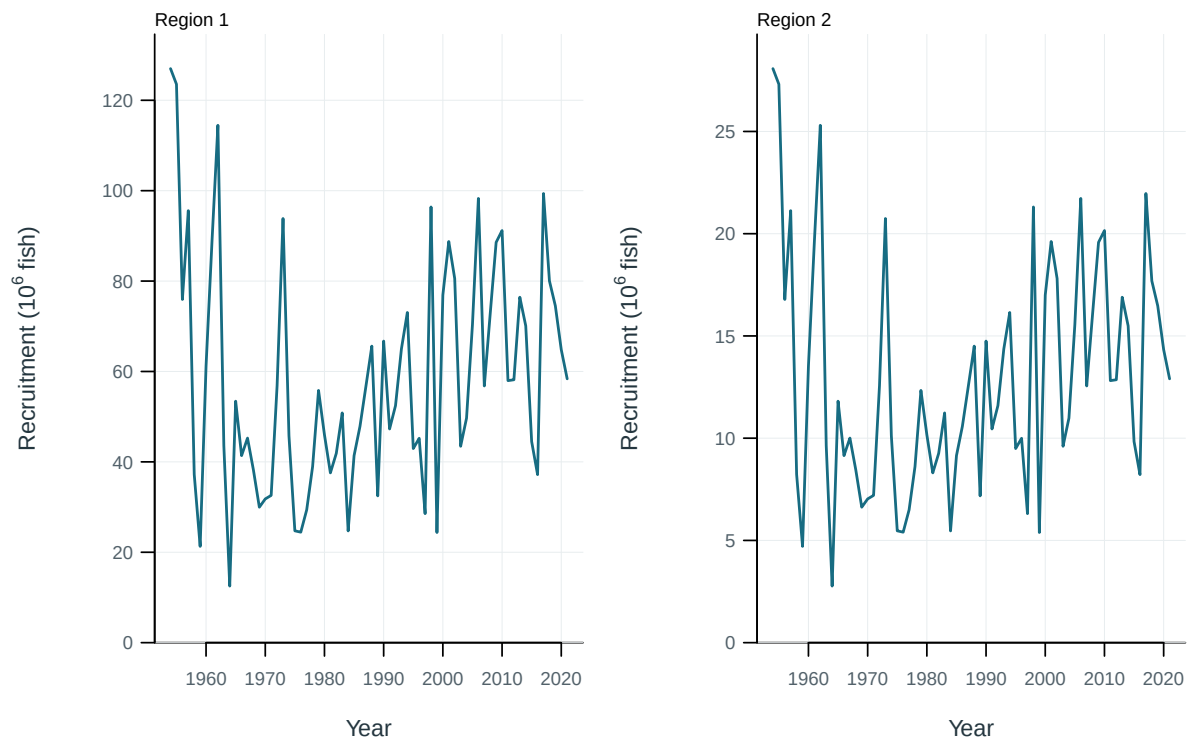


Figure 39: Recruitment by model region.

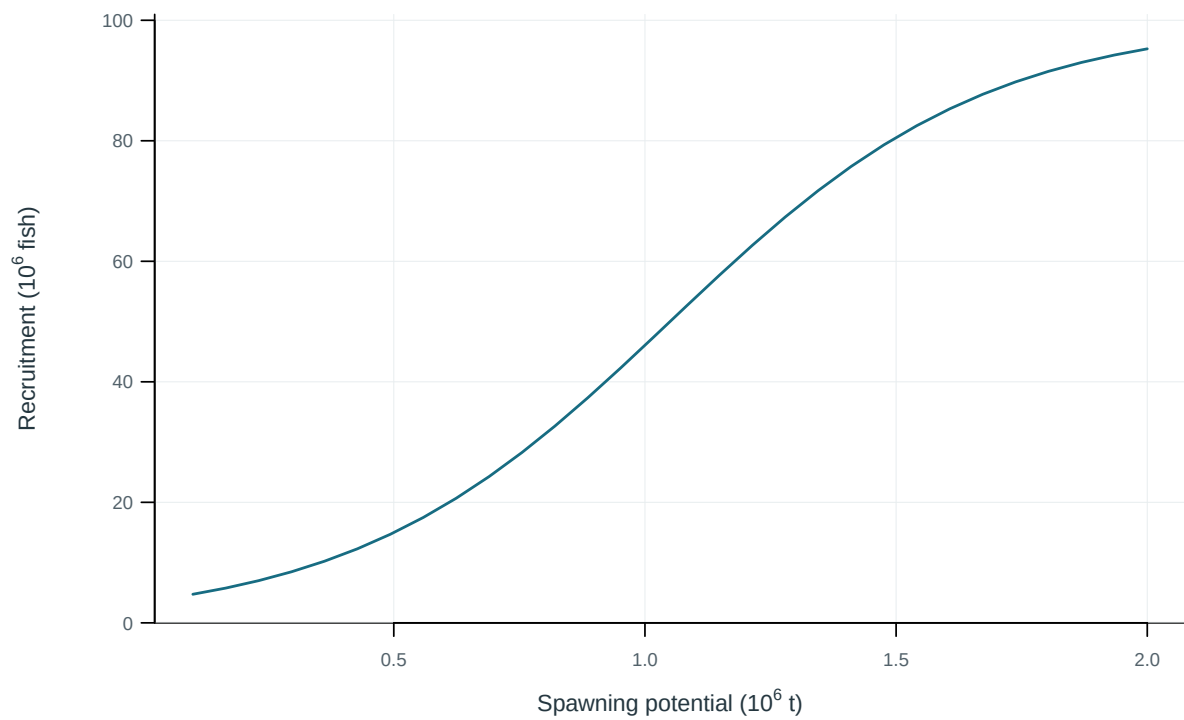


Figure 40: Illustrative output. Stock–recruitment observations and fitted relationship.

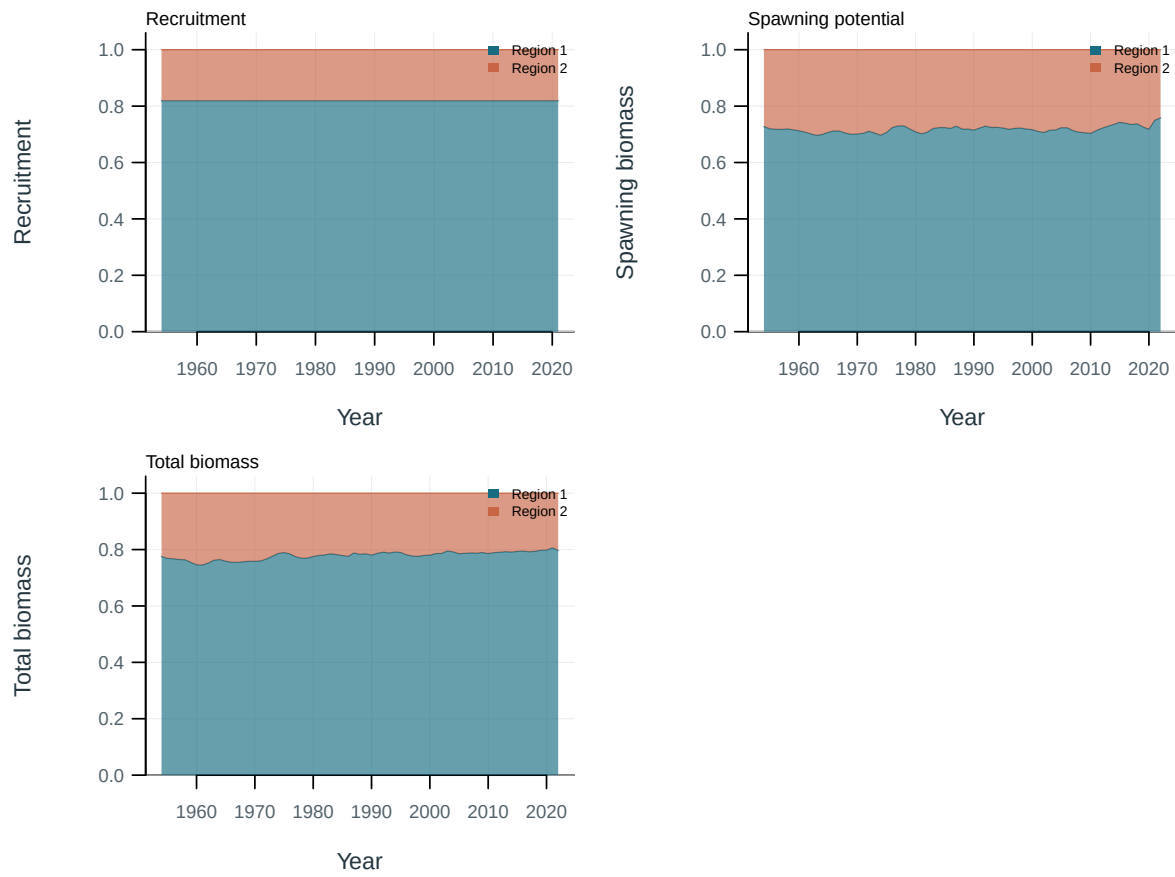


Figure 41: Regional shares of population quantities.

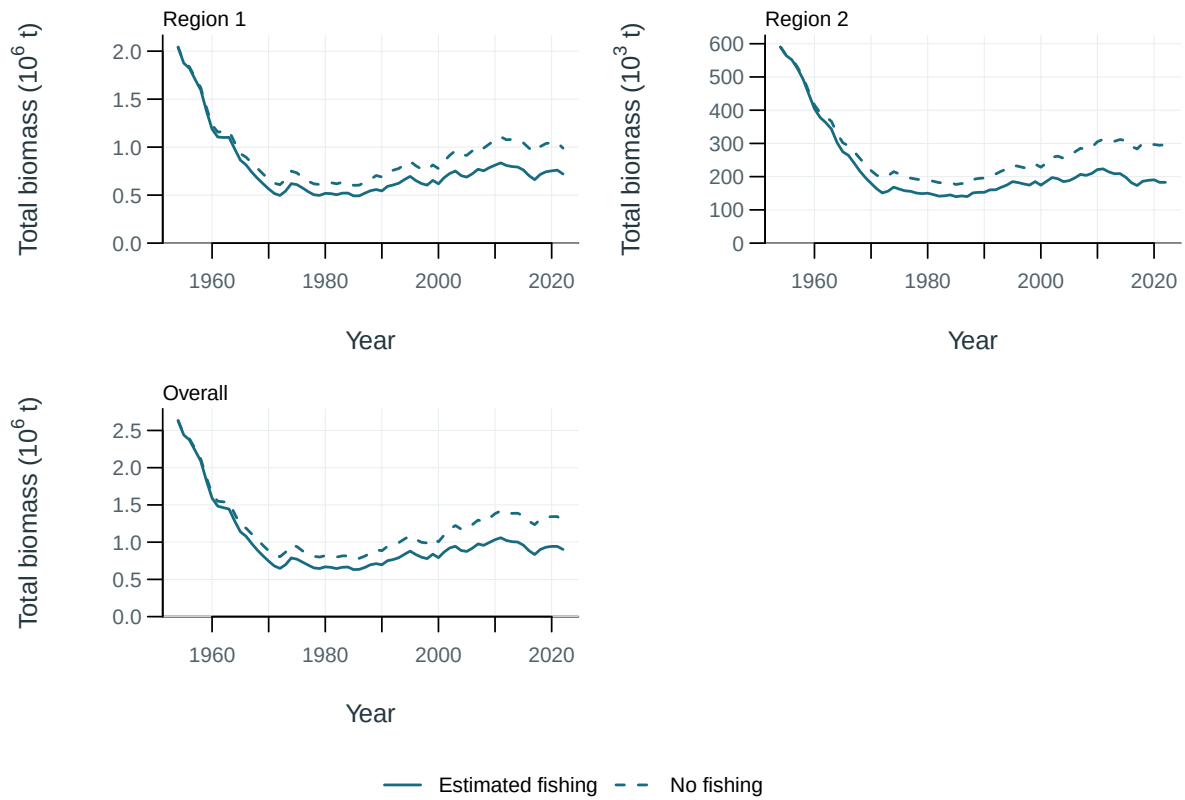


Figure 42: Total biomass under estimated fishing (solid) and the dynamic no-fishing counterfactual (dashed), by region and overall.

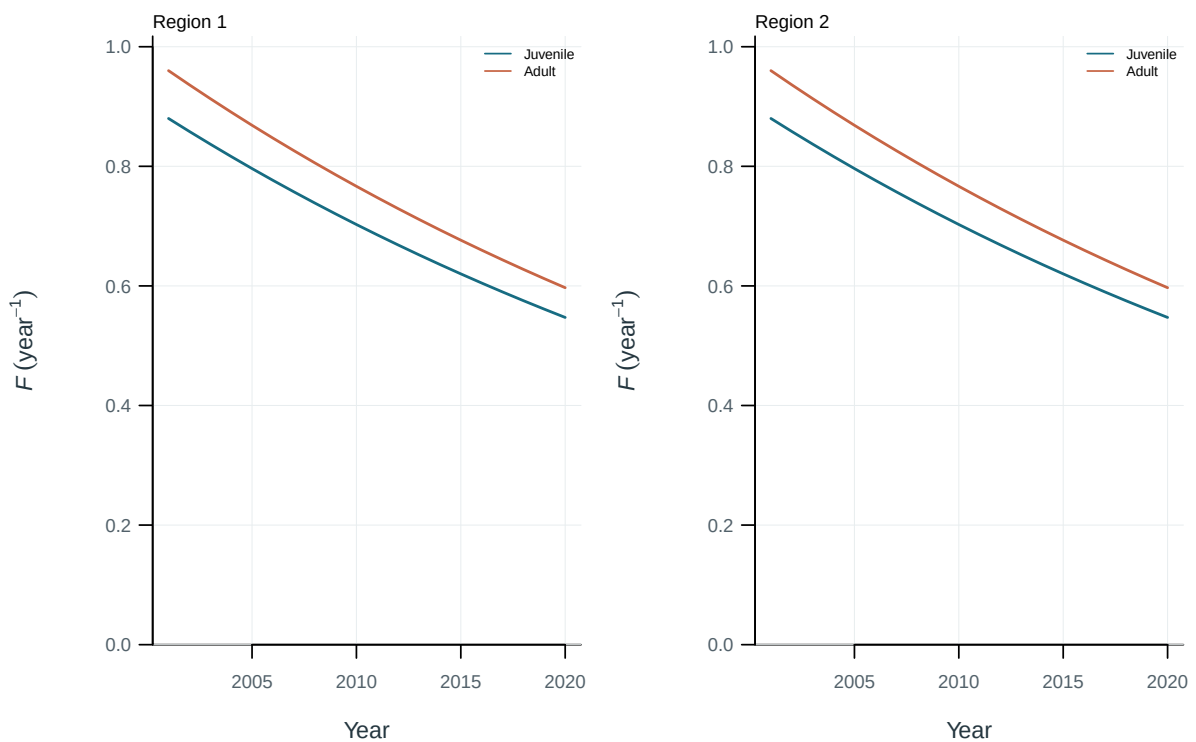


Figure 43: Illustrative output. Fishing mortality by life stage and region.

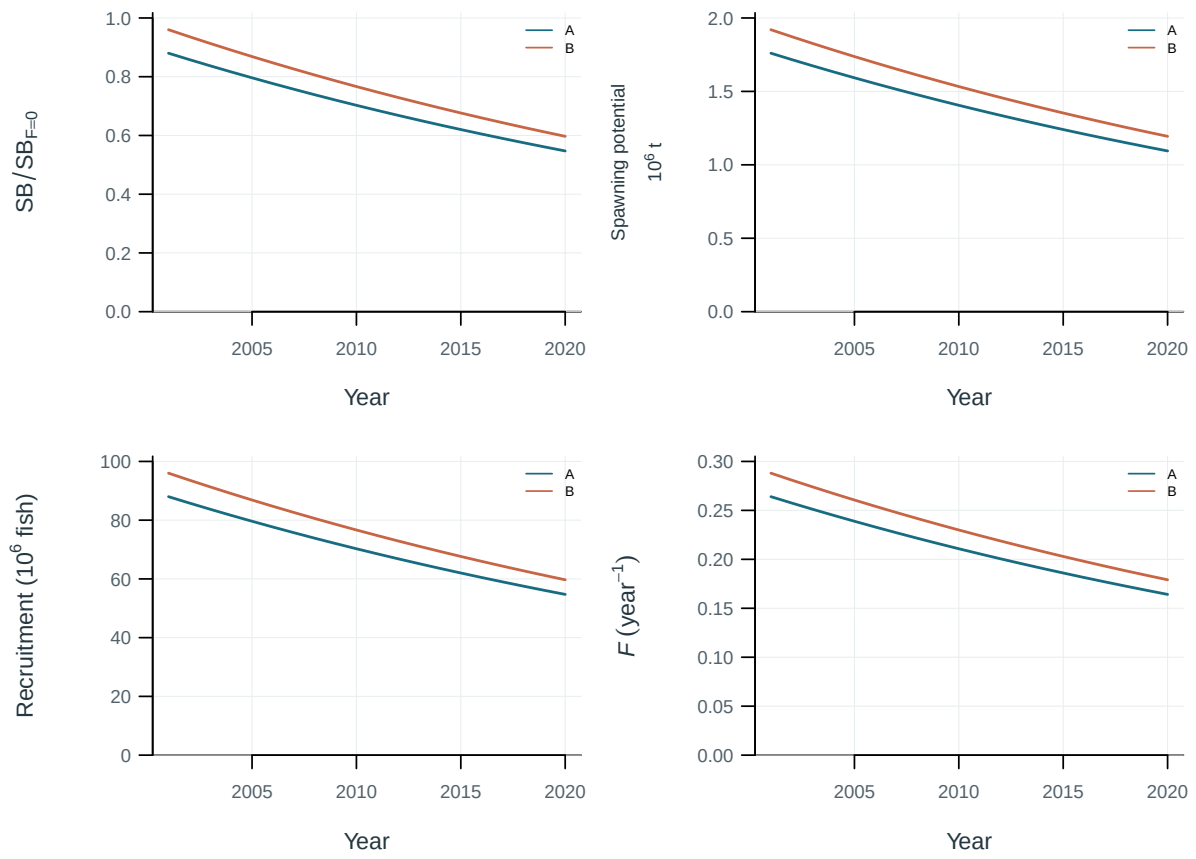


Figure 44: Sensitivity of population trajectories to steepness.

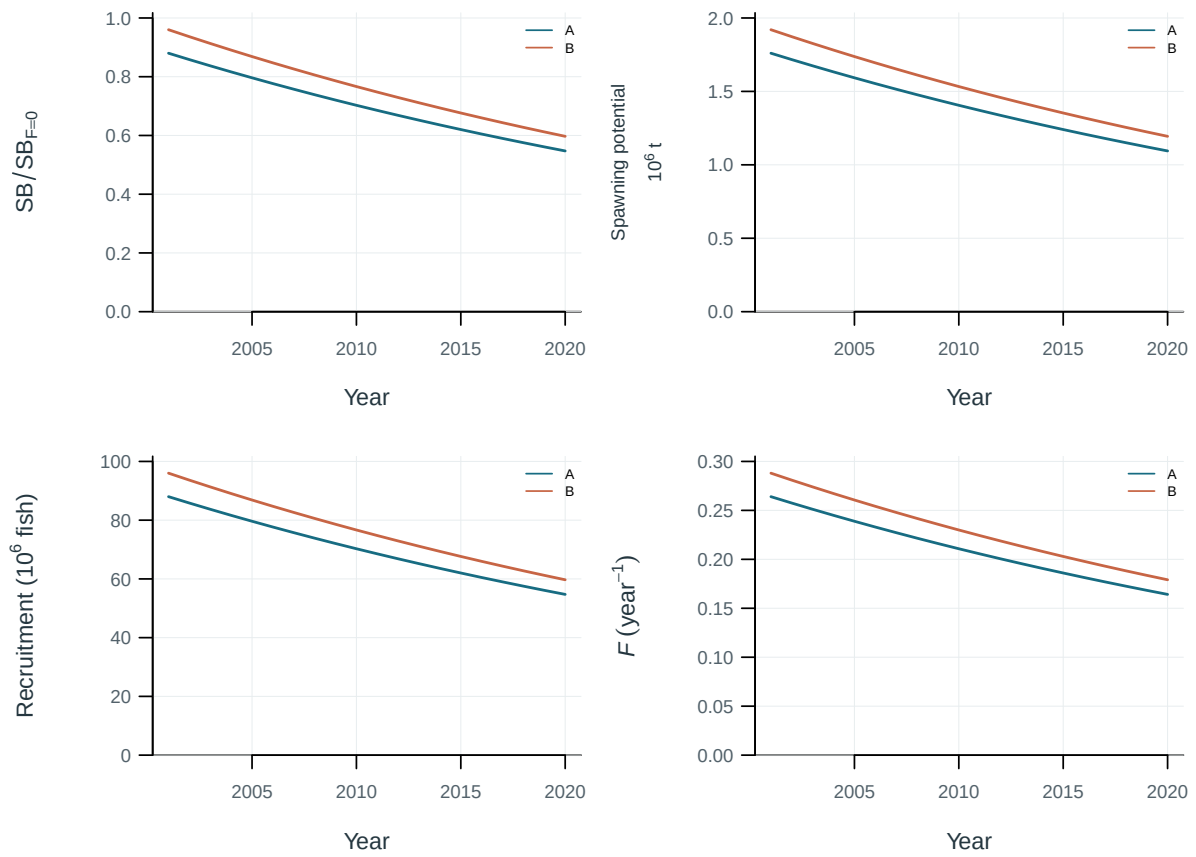


Figure 45: Sensitivity of population trajectories to natural mortality.

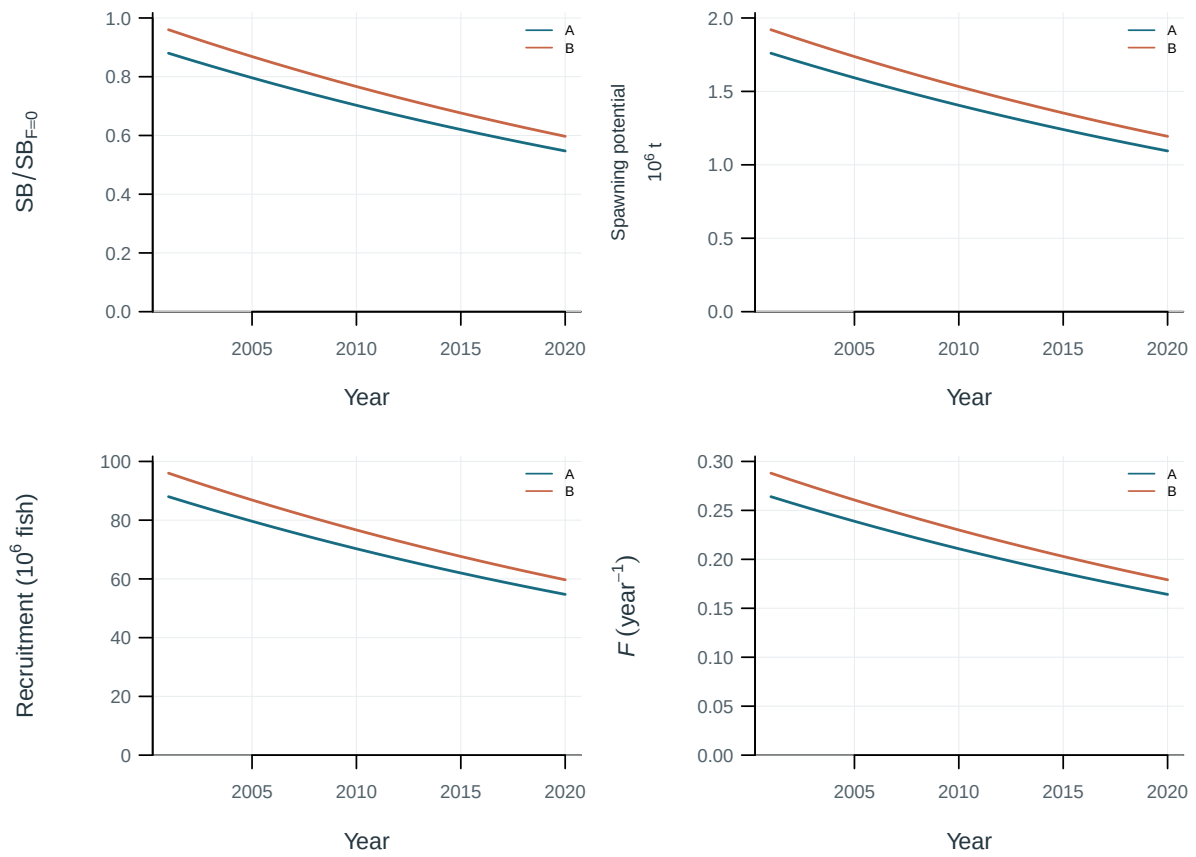


Figure 46: Sensitivity of population trajectories to regional scaling.

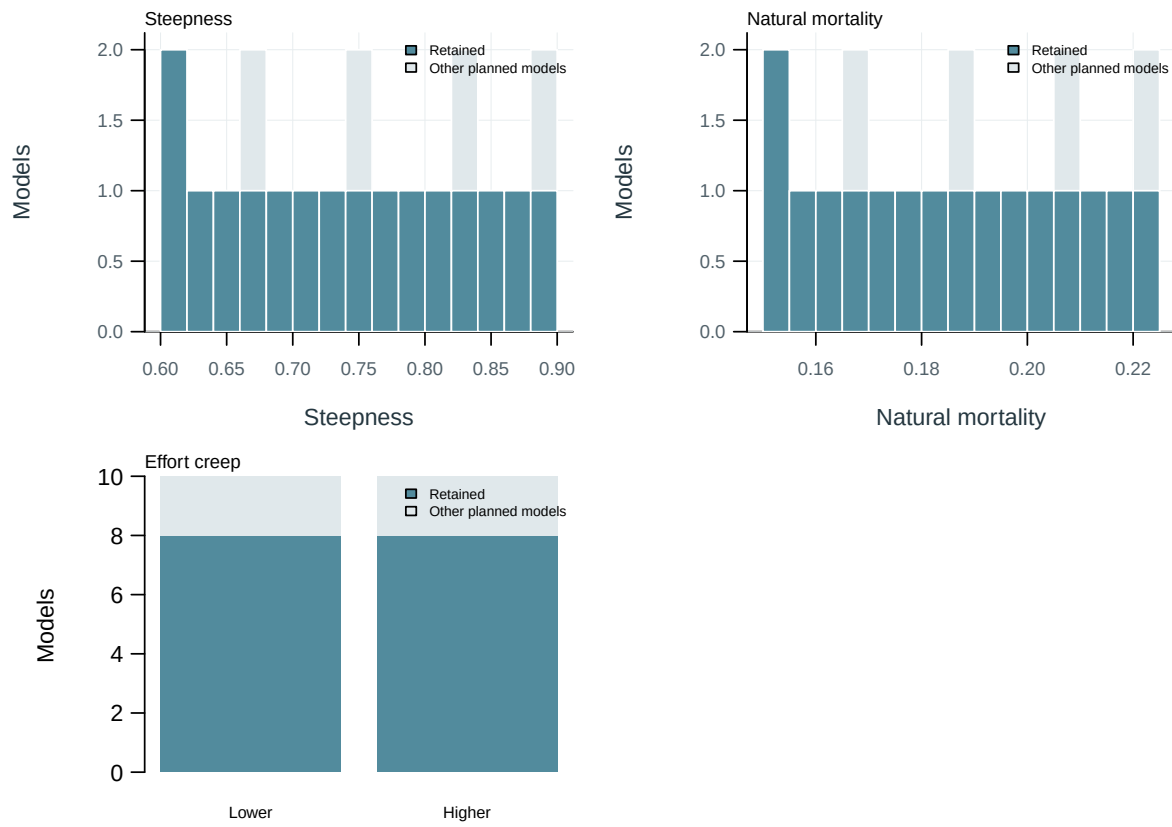


Figure 47: Planned and retained model inputs.

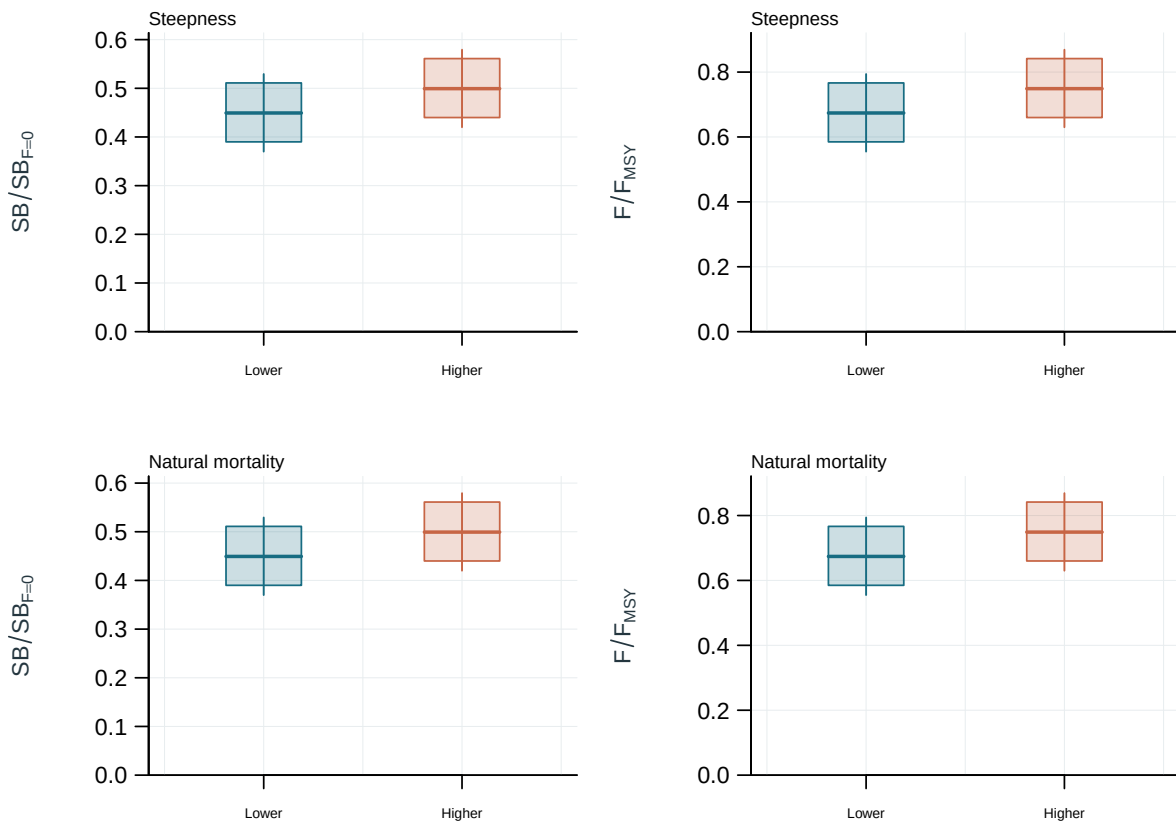


Figure 48: Weighted distributions of management quantities by design setting. Boxes: interquartile range; line: median; whiskers: 2.5th and 97.5th weighted percentiles.

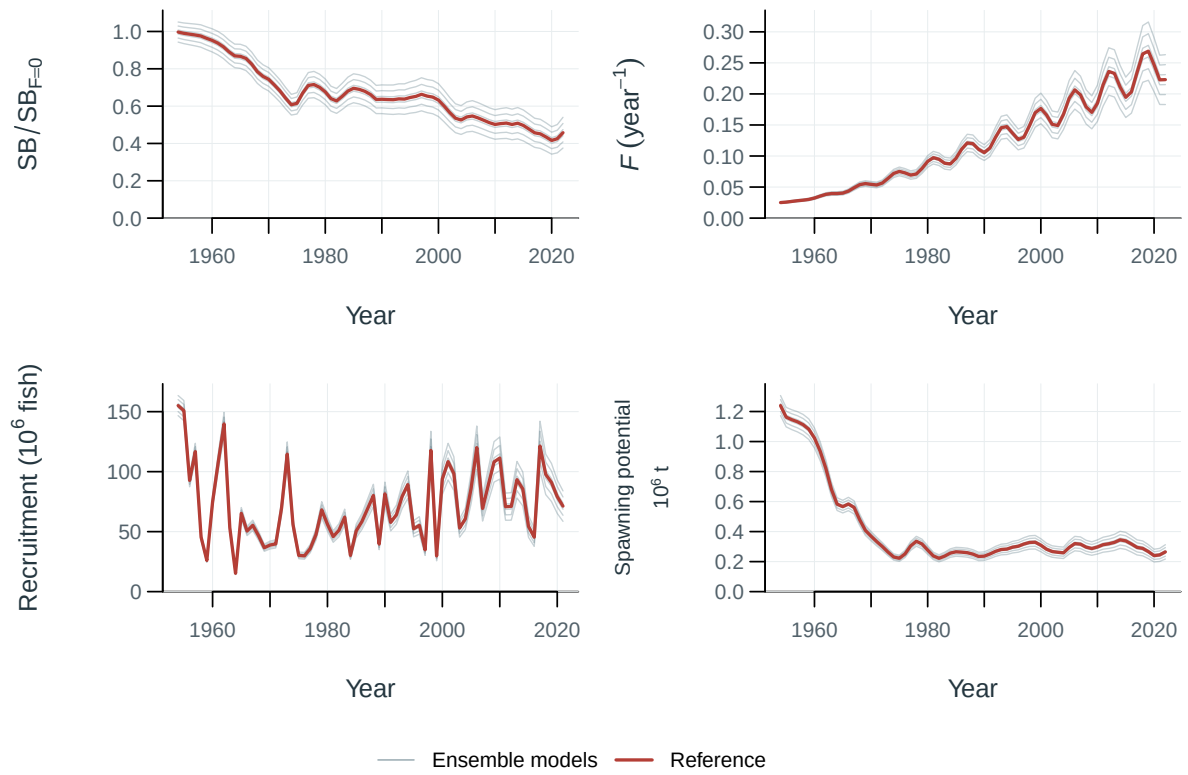


Figure 49: Population trajectories across ensemble models (thin lines) and the reference model (red).

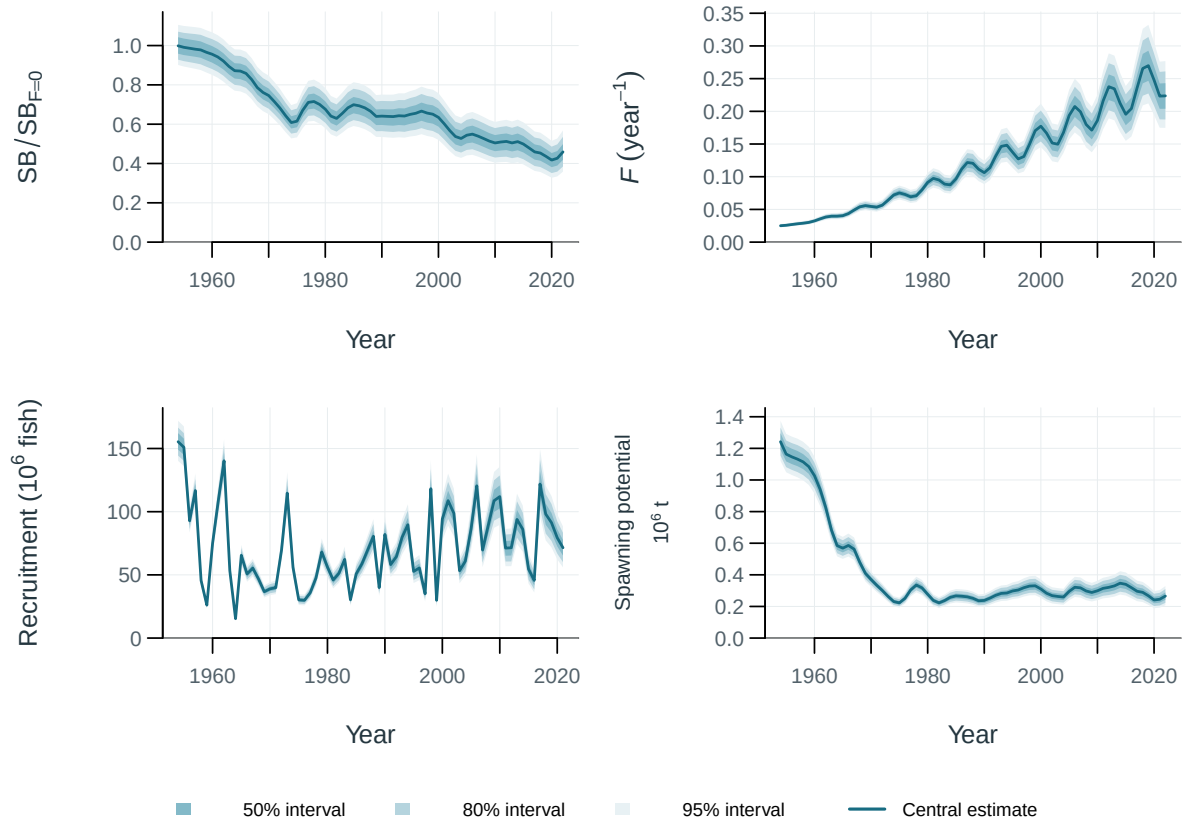


Figure 50: Population trajectories with supplied uncertainty intervals. Central 50, 80, 95% intervals (illustrative model spread and parameter perturbations).

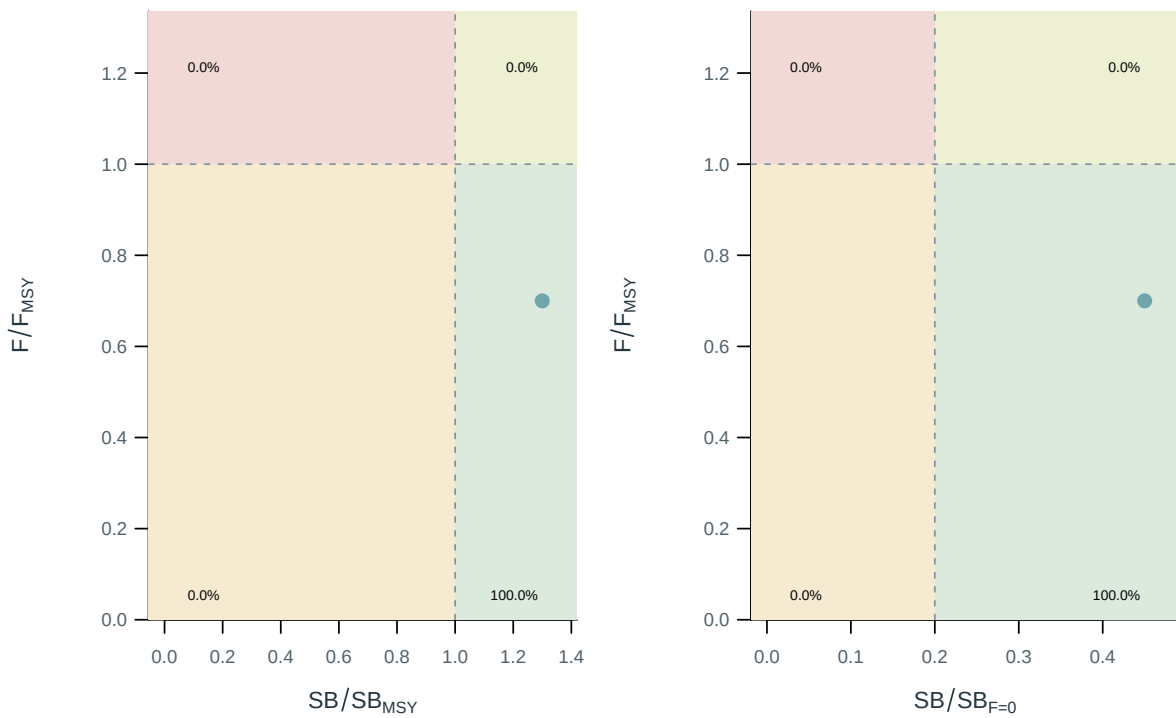


Figure 51: Stock status relative to the supplied spawning and fishing reference points.

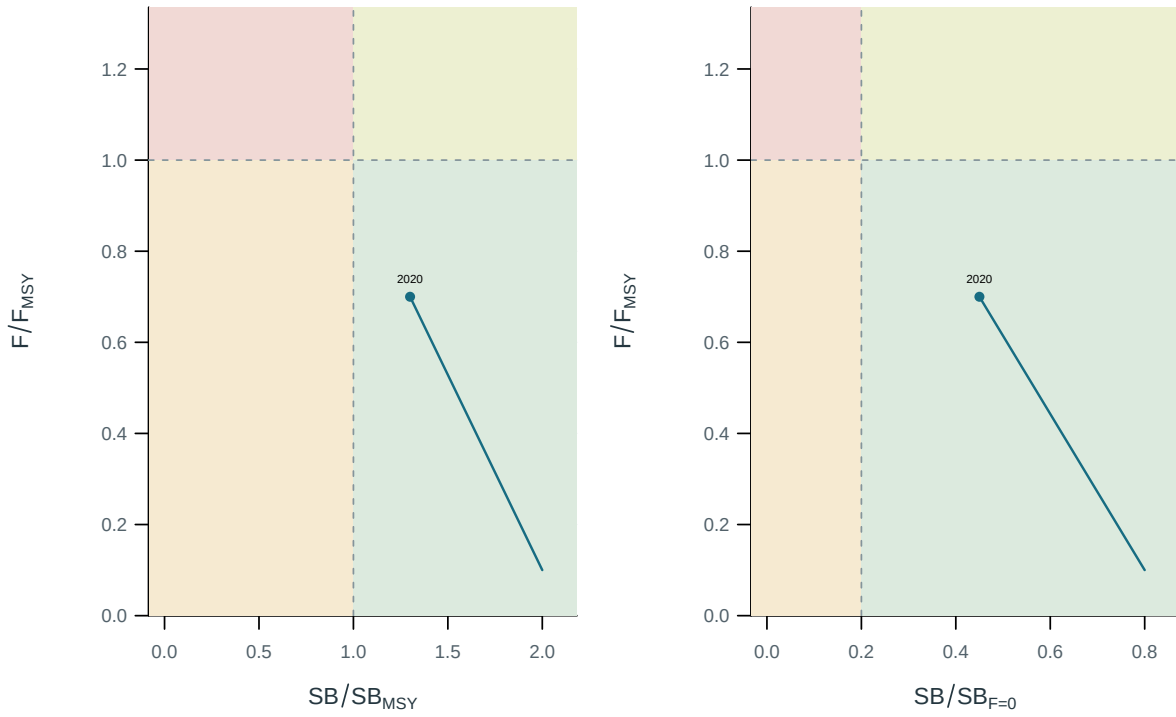


Figure 52: Stock status relative to the supplied spawning and fishing reference points.

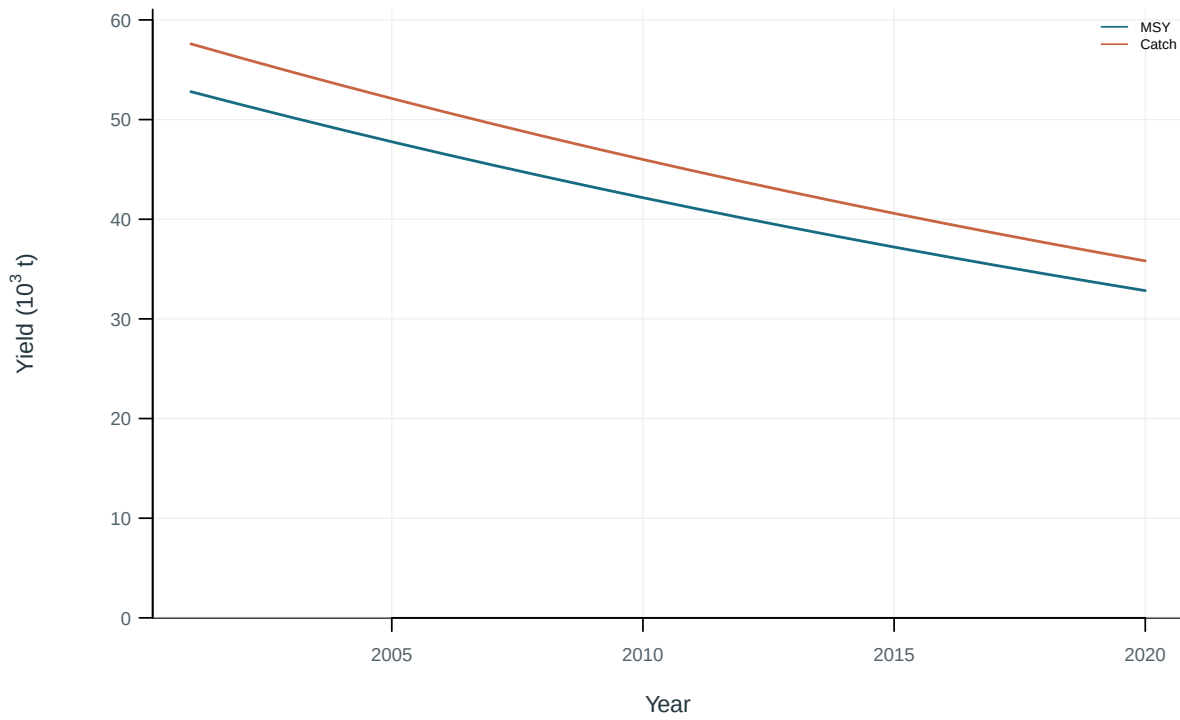


Figure 53: Dynamic maximum sustainable yield and catch.

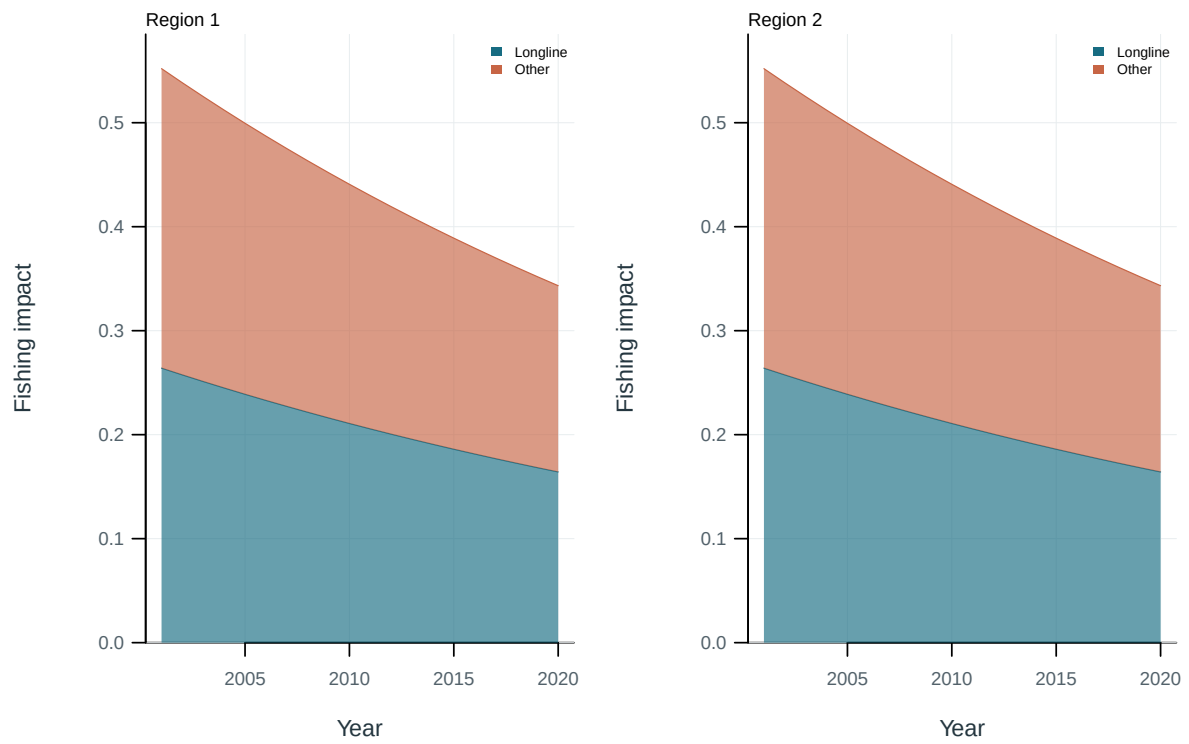


Figure 54: Illustrative output. Contributions of fishing groups to reduced spawning potential.

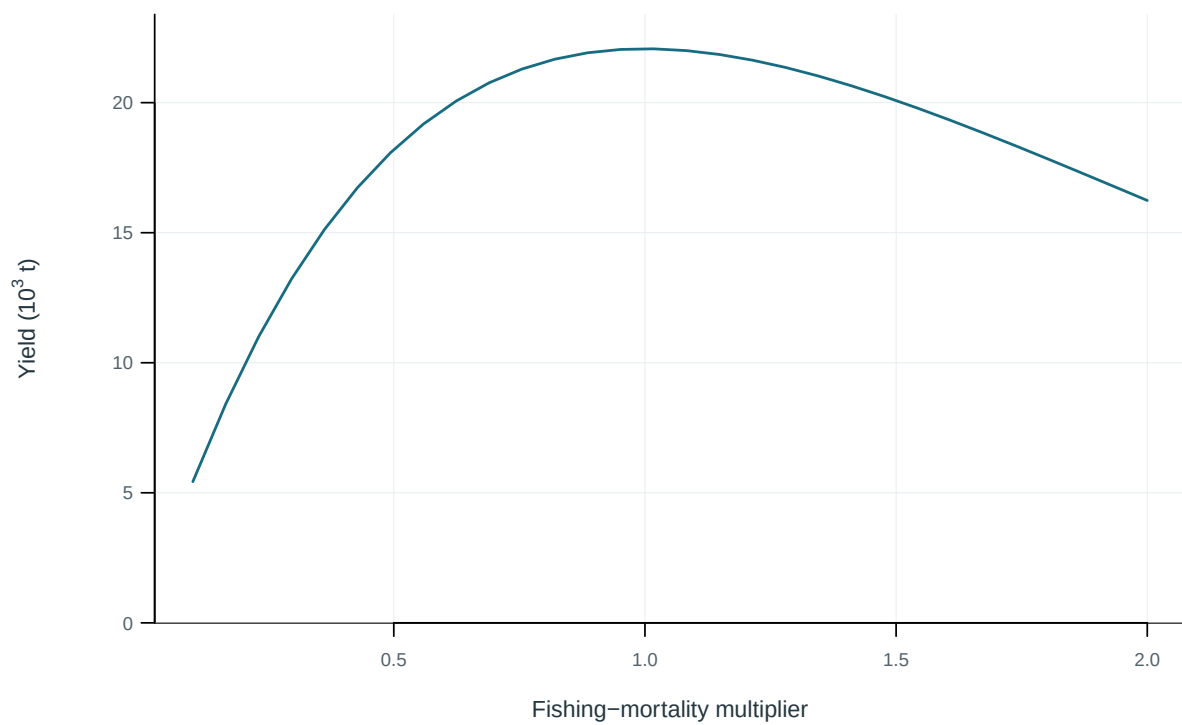


Figure 55: Equilibrium yield across fishing-mortality multipliers.

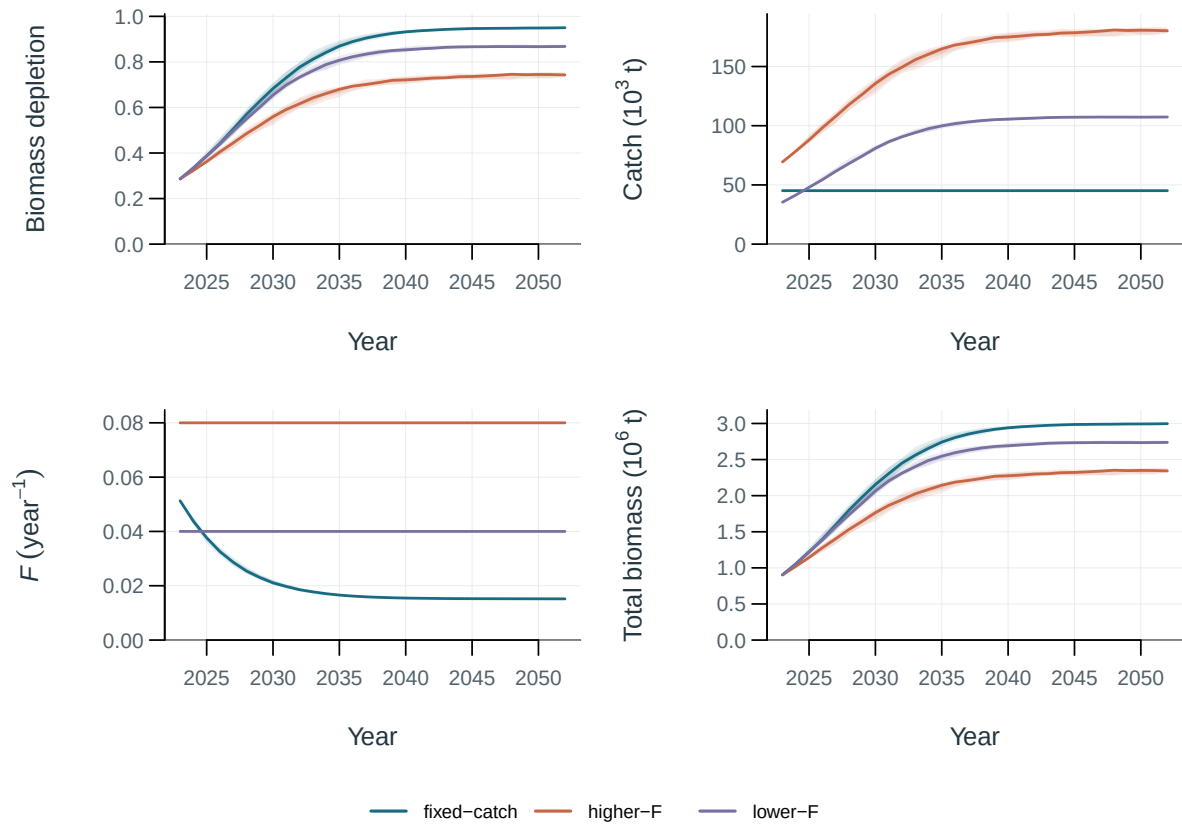


Figure 56: Illustrative surplus-production projections under fixed catch and two F scenarios. Lines are medians; bands span the central 50%, 80% and 95% of production-variability replicates.

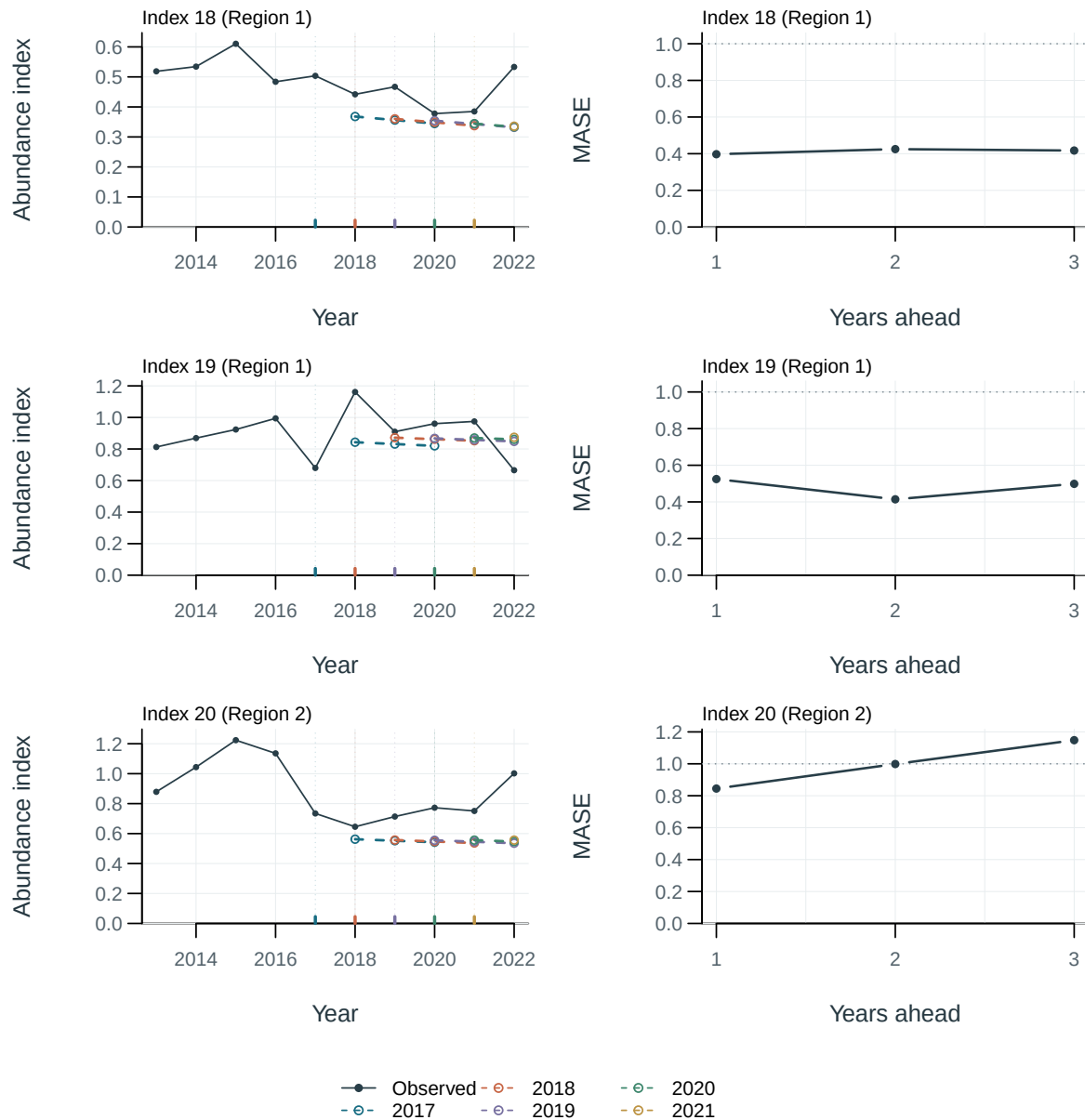


Figure 57: Observations (black) and held-out predictions (coloured dashed lines); vertical marks identify training cutoffs. Right panels show mean absolute scaled error (MASE) by forecast lead. The dotted line marks the training naive-error scale of one; lower values indicate smaller errors.

16 Supplementary outputs

[ALB · Diagnostic model](#)

[ALB · Diagnostic model figures and tables](#)

[ALB · Uncertainty ensemble](#)

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17 Appendices

17.1 Biological inputs and movement

[Document supporting growth, mortality and reproductive relationships and any external movement inputs. State units, constraints and parameter sources. Remove this subsection if the main methods provide sufficient detail.]